Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

WIND Thematic Assembly Centre
Regional High-Resolution WIND Product

WIND_ARC_PHY_HR_L3_NRT_012_100

WIND_ATL_PHY_HR_L3_NRT_012_101

WIND_BAL_PHY_HR_L3_NRT_012_102

WIND_BLK_PHY_HR_L3_NRT_012_103

WIND_MED_PHY_HR_L3_NRT_012_104

WIND_ARC_PHY_HR_L3_MY_012_105

WIND_ATL_PHY_HR_L3_MY_012_106

WIND_BAL_PHY_HR_L3_MY_012_107

WIND_BLK_PHY_HR_L3_MY_012_108

WIND_MED_PHY_HR_L3_MY_012_109

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ACRONYM TABLE

Acronym	Description
ARC	ARCtic Ocean
ASC	Ascending orbit
ATL	ATLantic Ocean (North-East part)
BAL	BALtic Sea
BLK	BLacK Sea
CLS	Collecte Localisation Satellite
CP	Cross-Polarisation
CYMS	CYclone Monitoring Seervice
DH	Dual-Horizontal acquisition mode, with both HH and HV channels
DSC	Descending orbit
DV	Dual-Vertical acquisition mode, with both VV and VH channels
ECMWF	European Center for Medium-range Weather Forecast
EIS	Entry Into Service
ESA	European Spatial Agency
EW	Extended Wide
GMF	Geophysical Model Function
HH channel	Channel emitting in Horizontal Polarization and receiving in Horizontal polarization
HR	High-Resolution
HV channel	Channel emitting in Horizontal Polarization and receiving in Vertical polarization
IFREMER	Institut Français pour la Recherche et l'Eploitation de la MER
IFS	Integrated Forecasting System (from ECMWF)
IPF	Instrument Processing Facility
IW	Interferometric Wide
L2	Level-2
L2-OCN	Level-2 OCeaN
L3	Level-3
MED	MEDiterranean Sea
MY	Multi-year
NRCS	Normalized Radar Cross Section
NRT	Near Real Time
NWP	Numerical Weather Prediction
PP	Co-polarisation
PR	Polarization Ratio
PU	Production Unit
PUM	Product User Manual
RMSD	Root Mean Square Deviation
S1A	Sentinel-1A
S1B	Sentinel-1B

Acronym	Description
S1C	Sentinel-1C
SAR	Synthetic Aperture Radar
SFMR	Stepped Frequency Microwave Radiometer
SI	Scatter index
SMAP	Soil Moisture Active and Passive mission
SMOS	Soil Moisture and Ocean Salinity
SNR	Signal-Noise Ratio
TAC	Copernicus Marine Service Thematic Assembly Centre
VH channel	Channel emitting in Vertical Polarization and receiving in Horizontal polarization
VV channel	Channel emitting in Vertical Polarization and receiving in Vertical polarization

KEY DEFINITIONS

co-pol channel	Channel where both the emitting and receiving polarization of the signal are the same : either both Vertical (VV channel), or both Horizontal (HH channel)
cross-pol channel	Acquisition channel where the receiving polarization is orthogonal to the emitting one; either emitting in Vertical polarization and receiving in Horizontal polarization (VH channel) or emitting in Horizontal polarization and receiving in Vertical polarization.
Dual-Vertical	Acquisition mode where both co-pol channel VV and cross-pol channel VH are received
Dual-Horizontal	Acquisition mode where both co-pol channel HH and cross-pol channel HV are received
co-pol inversion	Wind retrieval algorithm performed on the co-pol channel only
dual-pol inversion	Wind retrieval algorithm performed on both the co-pol and cross-pol channels (when available)

RELEVANT DOCUMENT LIST

[PUM 012_100]	Product User manual for WIND_ARC_PHY_HR_L3_NRT_012_100 https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-100.pdf
[PUM 012_101]	Product User manual for WIND_ATL_PHY_HR_L3_NRT_012_101 https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-101.pdf
[PUM 012_102]	Product User manual for WIND_BAL_PHY_HR_L3_NRT_012_102 https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-102.pdf
[PUM 012_103]	Product User manual for WIND_BLK_PHY_HR_L3_NRT_012_103 https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-103.pdf
[PUM 012_104]	Product User manual for WIND_MED_PHY_HR_L3_NRT_012_104 https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-104.pdf
[PUM 012_105]	Product User manual for WIND_ARC_PHY_HR_L3_MY_012_105 https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-100to109.pdf
[PUM 012_106]	Product User manual for WIND_ATL_PHY_HR_L3_MY_012_106 https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-100to109.pdf
[PUM 012_107]	Product User manual for WIND_BAL_PHY_HR_L3_MY_012_107 https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-100to109.pdf
[PUM 012_108]	Product User manual for WIND_BLK_PHY_HR_L3_MY_012_108 https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-100to109.pdf
[PUM 012_109]	Product User manual for WIND_MED_PHY_HR_L3_MY_012_109 https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-100to109.pdf
[PUM 012_10X]	Refers to all [PUM 012_100], [PUM 012_101], [PUM 012_102], [PUM 012_103], [PUM 012_104], [PUM 012_105], [PUM 012_106], [PUM 012_107], [PUM 012_108], and [PUM 012_109]

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the quality of the Near Real-Time (NRT) and multiyear (MY) reprocessed Level-3 (L3) High-Resolution (HR) wind products based on Sentinel-1A, Sentinel-1B and Sentinel-1C C-band Synthetic Aperture Radar (SAR) observations. The L3 NRT products are produced by Collecte Localisation Satellites (CLS) and the L3 MY products are generated by the French Institute for the SEA (IFREMER). Both are disseminated by the Copernicus Marine Service (Table 1).

The WIND_XXX_PHY_L3_NRT_012_00X and WIND_XXX_PHY_L3_MY_012_00X products where XXX and X represent the product basin and its associated digit, respectively, (see Table 1 for the complete list of products) contain daily gridded wind vectors obtained from Sentinel-1A, Sentinel-1B and Sentinel-1C C-band Synthetic Aperture Radar (SAR), Level 2 OCN (L2-OCN) observations. The L2-OCN observations used as inputs are made available by the European Space Agency (ESA).

The NRT production unit uses the latest ECMWF operational Integrated Forecasting System (IFS) forecast as *a priori* wind to process the files within 3 hours after the upstream release, while the MY production unit uses the ERA5 reanalysis as *a priori* wind.

Although L2-OCN products already contain wind vectors, those are calculated using the CMOD5.n Geophysical Model Function (GMF)[1] applied to the co-polarized (co-pol) VV channel (emitting in Vertical polarization and receiving in Vertical polarization). This VV GMF was mapped from scatterometer sensors (Hersbach et al., 2007), which are only able to use co-pol measurements. However, these co-pol GMF are known to lose sensitivity for wind above 20 m/s. Therefore, wind based on such GMF alone, are known to under-estimate wind speed (Polverari et al., 2022). For the L3 products winds based on SAR, we take advantage of the available cross-polarized (cross-pol) VH channel (emitting in Vertical polarization and receiving in Horizontal polarization) for which GMF were specifically derived based on C-Band SAR (Mouche et al., 2017, Mouche et al., 2019). Winds estimated from the combination of both the co-pol and cross-pol channels are referred to as dual-polarization (or dual-pol) winds.

As shown in Mouche et al. (2019), taking advantage of dual polarization strongly improves the wind estimation for high wind conditions, thanks to the much greater VH channel sensitivity compared to VV. The associated algorithm has also been implemented for the L3 production.

This document addresses the validation of the L3 winds derived using a SAR-based GMF (summarizing previous studies on the subject), and present a first evaluation of the dual-pol algorithm (instead of the co-pol only) under relatively moderate wind conditions, as opposed to cyclonic conditions.

[1] Described briefly in section II.4.1

Table 1: HR-Wind products: description and production unit

Product	Product description	Production Unit
WIND_ARC_PHY_HR_L3_NRT_012_100	Arctic Ocean HR wind L3 NRT	CLS
WIND_ATL_PHY_HR_L3_NRT_012_101	North-East Atlantic HR wind L3 NRT	CLS
WIND_BAL_PHY_HR_L3_NRT_012_102	Baltic Sea HR wind L3 NRT	CLS
WIND_BLK_PHY_HR_L3_NRT_012_103	Black Sea HR wind L3 NRT	CLS
WIND_MED_PHY_HR_L3_NRT_012_104	Mediterranean Sea HR wind L3 NRT	CLS
WIND_ARC_PHY_HR_L3_MY_012_105	Arctic Ocean HR wind L3 MY	Ifremer
WIND_ATL_PHY_HR_L3_MY_012_106	North-East Atlantic HR wind L3 MY	Ifremer
WIND_BAL_PHY_HR_L3_MY_012_107	Baltic Sea HR wind L3 MY	Ifremer
WIND_BLK_PHY_HR_L3_MY_012_108	Black Sea HR wind L3 MY	Ifremer
WIND_MED_PHY_HR_L3_MY_012_109	Mediterranean Sea HR wind L3 MY	Ifremer

I.2 Summary of the results

The quality of the High-Resolution Wind SAR L3 products relies heavily on the quality of the L2 products (hereafter referred to as L2-HRWIND). These L2 products are generated by the HR-WIND L2-to-L2 processing chain, which performs a dual-pol wind inversion. The dual-pol wind inversion is carried out in two steps: a co-pol and a cross-pol wind inversion (see Figure 1).

For Sentinel-1 products acquired in IW (Interferometric Wide), EW (Extended Wide) acquisition modes and in DV (Dual-Vertical VV/VH) polarization, the first step of the L2-to-L2 processing is identical to that routinely performed by the ESA S1 IPF (Sentinel-1 Instrument Processing Facility). The intermediate results (referred to as L2_co-pol) have been compared to the L2-OCN (outputs for IPF S1 processing) products to validate the nominal operation of the processing chain. The comparison between L2_co-pol and L2-OCN over one cycle (from 1/03/2023 to 12/03/2023) gives a bias of 1.35×10^{-5} m/s, with a standard deviation of 0.01 m/s, for the IW_DV and EW_DV acquisition modes. This result confirms the validity of the first step of the processing. For Sentinel-1 products acquired in IW (Interferometric Wide), EW (Extended Wide) acquisition modes and in DH (Dual-Horizontal HH/HV) polarization, the first step of the L2-to-L2 processing is different, since the HR-WIND processor uses Zhang2 (Zhang et al., 2011) polarization ratio (PR) while the ESA S1 IPF uses Mouche 2 PR.

The first step (co-pol inversion) is then followed by a second step (the cross-pol inversion), specific to the HR-WIND processor, necessary to obtain the L2-HRWIND products.

The quality of the L2-HRWIND products is assessed through comparison with NWP (Numerical Weather Prediction) models, remote sensing and in-situ data, a summary of which is presented in the following subsection.

The addition of the new Sentinel-1C satellite to the NRT products[2] is validated at the product level, i.e. L3, through verifying that the comparison of S1C L3 NRT products with NWP models gives similar results as the comparison of S1A L3 NRT products with the same NWP models.

A more detailed description of the validation performed is available in section IV.

[2] starting from the March 2026 Entry Into Service (EIS)

I.3 Estimated Accuracy Numbers

The performance of the HR WIND products is evaluated against three different wind datasets:

- An in-situ wind measurement dataset taken from the moored buoys of the CMEMS In-Situ TAC, to assess classical wind regimes around Europe.
- Numerical Weather Prediction (NWP) datasets (both ECMWF and NCEP models) to check the consistency of the wind inversion.
    - For the NRT stream these datasets are part of the L2-HRWIND validation over one S1A cycle (12 days in March 2024) and of the L3 joint-validation of S1C and S1C,
    - For MY, the dataset consists of the ECMWF IFS model for collocated measurements of S1A and S1B over 2018-2023.
- A HY2 scatterometer wind measurements dataset, co-located with Sentinel-1A.

The main results are displayed in Table 2 for the L2 validation of NRT, in Table 3 for the L3 validation of NRT and in Table 4 for MY.

In Table 2, the performance against Numerical Weather Predictions (NWP) models wind speeds is given for 2 different configurations of the antenna: IW-DV and EW-DH. The different performances for EW-DH (stronger bias and larger RMSD) than for IW-DV is related to the less favourable DH polarization and to the possible wrong masking of the sea ice coverage in the SAR image, leading to over-estimated wind speeds.

More details on the frameworks for validation (both NRT and MY) are given in Section V, while more comprehensive results are presented in Section VI.

Table 2: Performance of the L2-HRWIND NRT stream (S1A only) compared both to in-situ wind speed from co-located moorings and to co-located Numerical Weather Predictions (NWP) wind speed. For comparison with NWP, the performances are given according to the two main acquisition configurations (either IW-DV or EW-DH). The performances are given in terms of BIAS, root-mean-square difference (RMSD) and scatter index (SI).

L2 Metric for S1A	Acquisition configuration	Value	Units
WIND_spd-CLASS2-MOOR-BIAS	all	-1.67	m/s
WIND_spd-CLASS2-MOOR-RMSD	all	2.69	m/s
WIND_spd-CLASS2-MOOR-SI	all	21.48	%
WIND_spd-CLASS4-MOD_emcwf_ifs-BIAS-12D	IW-DV	0.30	m/s
WIND_spd-CLASS4-MOD_emcwf_ifs-RMSD-12D	IW-DV	1.64	m/s
WIND_spd-CLASS4-MOD_ncep-BIAS-12D	IW-DV	0.18	m/s
WIND_spd-CLASS4-MOD_ncep-RMSD-12D	IW-DV	1.85	m/s
WIND_spd-CLASS4-MOD_emcwf_ifs-BIAS-12D	EW-DH	1.56	m/s
WIND_spd-CLASS4-MOD_emcwf_ifs-RMSD-12D	EW-DH	3.11	m/s
WIND_spd-CLASS4-MOD_ncep-BIAS-12D	EW-DH	0.74	m/s
WIND_spd-CLASS4-MOD_ncep-RMSD-12D	EW-DH	2.19	m/s

Table 3 : Performance of the L3 NRT stream for S1A and S1C (for all acquisition configurations) compared to co-located Numerical Weather Predictions (NWP) wind speed. The performances are given in terms of BIAS, root-mean-square difference (RMSD) and scatter index (SI).

SAR Satellite	L3 Metric	Value	Units
S1A	WIND_spd-CLASS4-MOD_emcwf_ifs-BIAS-12D	0.20	m/s
S1A	WIND_spd-CLASS4-MOD_emcwf_ifs-RMSD-12D	3.38	m/s
S1A	WIND_spd-CLASS4-MOD_emcwf_ifs-SI-12D	0.55	1
S1A	WIND_spd-CLASS4-MOD_ncep-BIAS-12D	0.05	m/s
S1A	WIND_spd-CLASS4-MOD_ncep-RMSD-12D	3.30	m/s
S1A	WIND_spd-CLASS4-MOD_ncep-SI-12D	0.52	1
S1C	WIND_spd-CLASS4-MOD_emcwf_ifs-BIAS-12D	0.03	m/s
S1C	WIND_spd-CLASS4-MOD_emcwf_ifs-RMSD-12D	3.57	m/s
S1C	WIND_spd-CLASS4-MOD_emcwf_ifs-SI-12D	0.56	1
S1C	WIND_spd-CLASS4-MOD_ncep-BIAS-12D	-0.05	m/s
S1C	WIND_spd-CLASS4-MOD_ncep-RMSD-12D	3.51	m/s
S1C	WIND_spd-CLASS4-MOD_ncep-SI-12D	0.54	1

Table 4 : Performances (Bias, Root-Mean-Square Difference (RMSD) and Scatter Index (SI)) of the HR-WIND MY stream compared both to co-located HY2 scatterometer wind speeds (for S1A only) and to co-located Numerical Weather Predictions (NWP) wind speeds (for both S1A and S1B).

SAR Satellite	Metric	Value	Units
S1A	WIND_spd-CLASS4-L2SCAT_hy2-BIAS	0.12	m/s
	WIND_spd-CLASS4-L2SCAT_hy2-RMSD	1.40	m/s
	WIND_spd-CLASS4-L2SCAT_hy2-SI	14.78	%
S1A & S1B	WIND_spd-CLASS4-MOD_emcwf_ifs-BIAS	-0.22	m/s
	WIND_spd-CLASS4-MOD_emcwf_ifs-RMSD	3.93	m/s
	WIND_spd-CLASS4-MOD_emcwf_ifs-SI	42.22	%
	WIND_dir-CLASS4-MOD_emcwf_ifs-BIAS	0.03	degrees
	WIND_dir-CLASS4-MOD_emcwf_ifs-RMSD	60.06	degrees
	WIND_dir-CLASS4-MOD_emcwf_ifs-SI	29.92	%
S1A	WIND_spd-CLASS4-MOD_emcwf_ifs-BIAS	-0.37	m/s
	WIND_spd-CLASS4-MOD_emcwf_ifs-RMSD	4.34	m/s
	WIND_spd-CLASS4-MOD_emcwf_ifs-SI	45.97	%
	WIND_dir-CLASS4-MOD_emcwf_ifs-BIAS	-0.16	degrees
	WIND_dir-CLASS4-MOD_emcwf_ifs-RMSD	60.75	degrees
	WIND_dir-CLASS4-MOD_emcwf_ifs-SI	29.70	%
S1B	WIND_spd-CLASS4-MOD_emcwf_ifs-BIAS	0.08	m/s
	WIND_spd-CLASS4-MOD_emcwf_ifs-RMSD	3.00	m/s
	WIND_spd-CLASS4-MOD_emcwf_ifs-SI	33.12	%
	WIND_dir-CLASS4-MOD_emcwf_ifs-BIAS	0.39	degrees
	WIND_dir-CLASS4-MOD_emcwf_ifs-RMSD	58.73	degrees
	WIND_dir-CLASS4-MOD_emcwf_ifs-SI	30.37	%

II PRODUCTION SYSTEM DESCRIPTION

II.1 Production centre name

The HR-Wind Production Unit (PU) is responsible for providing L3 winds at high resolution (~1 km for the Arctic region and 0.01° for other CMEMS regions). These products are derived from Sentinel-1 SAR imagery and involve two production centres:

- CLS, which is in charge of the Near Real-Time (NRT) products
- IFREMER, which delivers the Multi-Year (MY) products

The derivation of the L3 wind products is based on the Level-2 OCN products computed and distributed by ESA, through several hubs[3].

II.2 Main principle of L2-to-L3 processing

Both NRT and MY production streams follow the same processing scheme (see Figure 1):

1) They ingest L2-OCN products (available on ESA hub), from which they read the Normalized Radar Cross Section (NRCS, also known as σ_0) in both VV and VH polarisations (when available)
2) They take an *a priori* wind from a NWP model:
   - For the NRT stream: the latest forecast from the ECMWF Integrated Forecasting System (IFS) that can be found in L2-OCN products as legacy of AUX_WND file from L1.
   - For the MY stream: ERA5 model.
3) They perform a first wind inversion based on the co-polarised NRCS with the CMOD5N GMF and *a priori* wind taken from the model (step 2).
4) They perform a second wind inversion based on the cross-polarised NRCS by minimizing a cost function, depending on the Geophysical Model Function MS1AHW2 and the *a priori* wind taken from step 3.
5) The slices are then re-gridded and concatenated on a daily basis.
6) Finally, the L2 flags are adapted towards L3 flags and the file is formatted according to CMEMS Guidelines (see Product User Manual, PUM [PUM 012_10X]).

[3] e.g. https://browser.dataspace.copernicus.eu/ or https://search.asf.alaska.edu

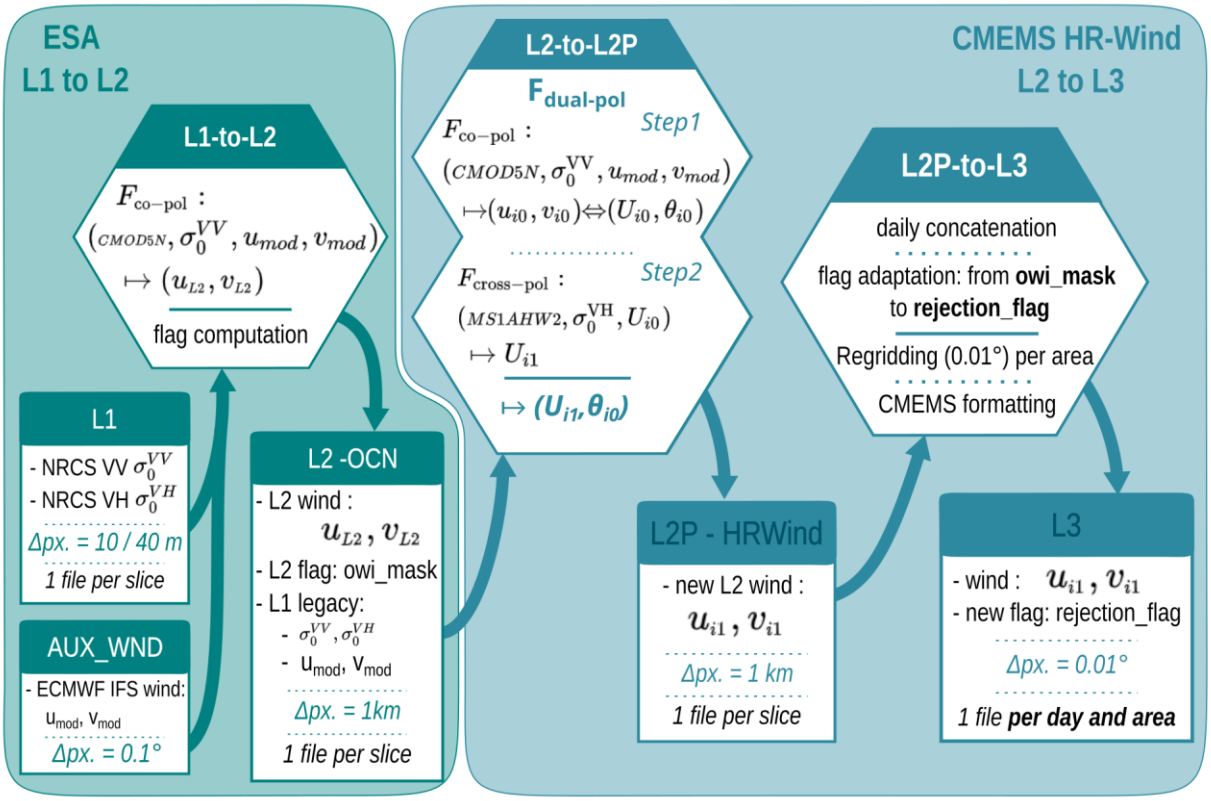

Figure 1: L3 wind processing flowchart.

II.3 Flags meanings

In relation to quality level of the data, in the HR-Wind L3 products there are three flags provided for each pixel (or grid cell):

1. quality_level: summarises the quality of each grid cell: “no_data”, “bad”, ”suspect”, “acceptable”, “good”.

2. rejection_flag: contains the reasons for any quality level different from “good”, coded onto the following bits (value is 1 if statement is TRUE, 0 if it is FALSE) :

- “some_portion_of_wvc_is_over_land”
- “some_portion_of_wvc_is_over_ice”
- “not_enough_good_sigma0_for_wind_retrieval”
- “bright_target_percentage_over_threshold” (threshold = 10%).
- “wind_inversion_not_successful”
- “cosmetic_pixel” (from format version 1.1): pixel added for visualisation by interpolating surrounding pixels (through nearest neighbour analysis) when the new variable “number_of_pixels” equals zero.

3. product_info: contains the acquisition and processing information (coded onto bits) for each pixel:

- “iw_acq_mode”: TRUE if Interferometric Wide (IW) mode,
- “ew_acq_mode”: TRUE if Extended Wide (EW) mode,
- “v_emitting_pol”: TRUE if vertical polarization emission,

- "h_emitting_pol": TRUE if horizontal polarization emission
- "single_pol_receiving": TRUE if single polarization acquisition,
- "dual_pol_receiving": TRUE if dual polarization acquisition,
- "single_pol_wind_processing": TRUE if co-pol algorithm is used,

"dual_pol_wind_processing": TRUE if dual-pol algorithm is used

Table 5 relates the "rejection_flag" bits of the L3 product to the flags in the L2 OCN (Ocean) product , from which they were inherited.

Table 6 gives the corresponding "rejection_flag" combinations for each "quality_level".

Table 5: Correspondence between L2-OCN product's parameters and L3 flags.

Status in input L2 OCN product	Bit in L3 product's rejection_flag
owiMask == 1	some_portion_of_wvc_is_over_land
owiMask == 2	some_portion_of_wvc_is_over_ice
owiMask == 4	not_enough_good_sigma0_for_wind_retrieval
owiPbright >10 %	bright_target_percentage_over_threshold

Table 6 : Correspondence betweenL3 product "rejection_flag" and L3 product "quality_level". The value of owiPbright is taken from the input L2-OCN files.

L3 product Rejection_flag	Associated quality_level
IF "not_enough_good_sigma0_for_wind_retrieval"	"no_data"
IF "wind_inversion_not_successful"	"no_data"
IF "cosmetic_pixel"	"no_data"
IF "some_portion_of_wvc_is_over_land"	"bad"
IF "some_portion_of_wvc_is_over_ice"	"bad"
IF "bright_target_percentage_over_threshold"	"suspect"
IF NOT "bright_target_percentage_over_threshold" AND owiPbright != 0 AND NOT "not_enough_good_sigma0_for_wind_retrieval" AND NOT "wind_inversion_not_successful" AND NOT "some_portion_of_wvc_is_over_land" AND NOT "some_portion_of_wvc_is_over_ice"	"acceptable_data"
IF owiPbright == 0 AND NOT "not_enough_good_sigma0_for_wind_retrieval" AND NOT "wind_inversion_not_successful" AND NOT "some_portion_of_wvc_is_over_land" AND NOT "some_portion_of_wvc_is_over_ice"	"Good"

II.4 Classical co-polarization GMF and Polarization Ratios

In this section, we present the coefficients for the CMOD5.n relationship between the radar backscatter (in VV polarization) and the wind speed and direction. We also present two different polarization ratios that are used to apply the CMOD algorithms to images acquired at HH polarization.

II.4.1 CMOD5.n co-polarization GMF

The CMOD5 model formulation is given in the appendix of the ECMWF Technical Memorandum No. 395 (Hersbach, 2003)[4].

For the CMOD5.n version, the coefficients have been updated in 2008 (Verhoef et al., 2008) and are shown in Table 7.

Table 7 : CMOD5.n coefficients

Coeff	Value (CMOD5.n)	Coeff	Value (CMOD5.n)	Coeff	Value (CMOD5.n)	Coeff	Value (CMOD5.n)
c1	-0.6878	*c8*	0.0159	*c15*	0.0066	*c22*	-3.3428
c2	-0.7957	*c9*	6.7329	*c16*	0.3222	*c23*	1.3236
c3	0.338	*c10*	2.7713	*c17*	0.012	*c24*	6.2437
c4	-0.1728	*c11*	-2.2885	*c18*	22.7	*c25*	2.3893
c5	0.0	*c12*	0.4971	*c19*	2.0813	*c26*	0.3249
c6	0.004	*c13*	-0.725	*c20*	3.0	*c27*	4.159
c7	0.1103	*c14*	0.045	*c21*	8.3659	*c28*	1.693

II.4.2 Polarization ratios

A polarization ratio (PR) is defined as the ratio (in linear units) between the NRCS values in VV and HH polarizations respectively

$$PR = \frac{\sigma_0^{VV}}{\sigma_0^{HH}} \tag{1}$$

II.4.2.1 Mouche2 polarization ratio

The PR known as Mouche2 has been defined in Mouche et al. (2005) and is a function of the incidence angle θ only. Its notation, as follows:

$$PR = A \exp(B\,\theta) + C \tag{2}$$

Where A, B and C are constants (A = 0.00799793; B = 0.125465 and C = 0.997379).

4 available at https://www.ecmwf.int/sites/default/files/elibrary/2003/9861-cmod5-improved-geophysical-model-function-ers-c-band-scatterometry.pdf.

II.4.2.2 Zhang2 Polarization ratio

The PR known as Zhang2 has been defined in Zhang et al. (2011) and is a function of both the incidence angle θ and the wind speed U_{10}.

$$PR = P(\theta)U_{10}^{Q(\theta)} \tag{3}$$

Where

$$P(\theta) = P1 \times \theta^2 + P2 \times \theta + P3$$

$$Q(\theta) = Q1 \times \theta + Q2$$

See Table 8 below for details of the fitted values for the coefficients (P1, P2 and P3; Q1 and Q2) applied.

Table 8 : Zhang2 polarization ratio coefficients

Coeff	P1	P2	P3	Q1	Q2
Fitted value (Zhang2)	0.0014	-0.0319	1.3794	0.0026	-0.1711

II.5 Processing chain genesis – High-winds framework

C-band SAR data are routinely processed by ESA to generate wind products (see Figure 1, upper part). However, these products only rely on the conventional Co-Polarisation (PP – stands for VV: Vertical-Vertical or HH: Horizontal-Horizontal) acquisitions. For extreme weather events, the Cross-Polarisation (CP – either HV or VH) exhibits a higher sensitivity to extreme wind conditions than the PP signal (Zhang & Perrie 2012) and it also shows less dependence on incidence angles and wind direction. Therefore, wind inversion methods have been developed to exploit CP signals and combine them with PP signals (Portabella et al., 2002, Zhang & Perrie 2012, Mouche et al., 2017). It should be noted that for more moderate winds, the signal-to-noise ratio of the CP signal is too low to be used exclusively (hence the combination with PP).

II.5.1 Dual-pol algorithm

The aim of the wind inversion is to minimize the following cost function:

$$J(u_{10}, v_{10}) = \left[\frac{u_{10}^{a\,priori} - u_{10}}{\Delta u_{10}}\right]^2 + \left[\frac{v_{10}^{a\,priori} - v_{10}}{\Delta v_{10}}\right]^2 + \sum_{p\in\{VV,VH\}} \left[\frac{\sigma_0^{pp} - GMF^{pp}(\theta,\phi,U_{10})}{\Delta\sigma_0^{pp}}\right]^2 \tag{4}$$

Where:

- (u_{10}, v_{10}) is the space of wind solutions,
- $(\Delta u_{10}, \Delta v_{10})$ their associated errors (which can be viewed as weights for the two first terms of the cost function),
- $(u_{10}^{\text{a priori}}, v_{10}^{\text{a priori}})$ is the a priori solution (classically given by a wind model forecast),
- σ_0^{pp} is the NRCS measurement and $\Delta\sigma_0^{pp}$ its associated error (which can be viewed as weights for the last terms of the cost function),
- $GMF^{pp}(\theta, \pi\ U_{10})$ is the Geophysical Model Function that links the NRCS to the wind speed while accounting for the incidence angle θ and the wind direction relation to antenna ϕ.

It should be noted that the equation is the same for the co-pol algorithm routinely implemented by ESA except that the last sum only contains the term with pp = VV or pp = HH.

In practice, for the dual-pol algorithm, the minimization of $J(u_{10}, v_{10})$ is not performed directly due to computation time considerations. Instead, the cost function J is separated into two cost functions J^{VV} and J^{VH} and whose minimizations are serialized.

$$J^{VV}(u_{10}, v_{10}) = \left[\frac{u_{10}^{\text{a priori}} - u_{10}}{\Delta u_{10}}\right]^2 + \left[\frac{v_{10}^{\text{a priori}} - v_{10}}{\Delta v_{10}}\right]^2 + \left[\frac{\sigma_0^{\text{VV}} - GMF^{VV}(\theta, \phi, U_{10})}{\Delta \sigma_0^{VV}}\right]^2 \quad (5.a)$$

$$J^{VH}(u_{10}, v_{10}) = \left[\frac{U_{10}^{\text{a priori}} - U_{10}}{\Delta U_{10}}\right]^2 + \left[\frac{\sigma_0^{\text{VH}} - GMF^{VH}(\theta, U_{10})}{\Delta \sigma_0^{VH}}\right]^2 \quad (5.b)$$

First, the minimization of the J^{VV} function is performed with $\left(u_{10}^{\text{a priori}}, v_{10}^{\text{a priori}}\right) = (u_{mod}, v_{mod})$ and $GMF^{VV} = CMOD5.N$ giving a first wind guess (u_{i0}, v_{i0}) or (U_{i0}, θ_{i0}) in polar coordinates.

Then the J^{VH} function is minimized with $U_{10}^{\text{a priori}} = U_{i0}$ and $GMF^{VH} = MS1AHW2$, giving the dual-pol wind estimate.

II.5.2 Cross-Polarization GMF (MS1AHW2)

As first noted in Mouche et al. (2017), the classical Cross Polarization (CP) GMFs were insufficient to represent wind speeds at high wind conditions and over all ranges of incidence angles observed by SAR (20° to 45°). For example, the CP GMF "H14S" saturates at wind speed higher than 35 m/s and incidence angles lower than 45°. The authors used a catalogue of SAR imageries to define a new GMF for CP more adapted to high wind speeds. Due to the ever-growing size of this catalogue of SAR imageries, the GMF has been refined in various instances (Mouche et al., 2019, internal CYMS[5] project communication). Here the GMF used is the latest version (MS1AHW2, independent of the wind direction) which is measured as follows:

$$\sigma_0 = GMF(\theta, U_{10}) = \frac{a_{Z1} U_{10}^{b_{Z1}}}{1 + \exp(-c_0(U_{10} - c_1))} + \frac{a_{Z2} U_{10}^{b_{Z2}}}{1 + \exp(-c_2(U_{10} - c_3))} \quad (6)$$

With

$$a_{Z1} = a_{0,Z1} \quad (6.a)$$

$$b_{Z1} = b_{0,Z1} + b_{1,Z1}\,\theta \quad (6.b)$$

$$a_{Z2} = a_{0,Z2} + a_{1,Z2}\,\theta + a_{2,Z2}\,\theta^2 \quad (6.c)$$

$$b_{Z2} = b_{0,Z2} + b_{1,Z2}\,\theta + b_{2,Z2}\,\theta^2 \quad (6.a)$$

[5] Cyclone Monitoring Service Project: https://www.esa-cyms.org/

Where $a_{i,Z1}, b_{i,Z1}, a_{i,Z2}, b_{i,Z2}, c_i \quad (i \in \mathbb{N})$ are constants empirically defined and shown in Table 9.

Table 9 : Coefficients for MS1AHW2 GMF (see equation 6)

Coeff	Value	Coeff	Value	Coeff	Value
$a_{0,Z1}$	2.13755392e-06	$a_{2,Z2}$	2.87698338e-08	c_0	-0.23257086
$b_{0,Z1}$	2.47395267	$b_{0,Z2}$	1.14509104e+00	c_1	12.39717002
$b_{1,Z1}$	-2.85775085e-03	$b_{1,Z2}$	3.41828829e-02	c_2	0.21667263
$a_{0,Z2}$	6.54058552e-05	$b_{2,Z2}$	-4.79715441e-04	c_3	12.22862991
$a_{1,Z2}$	-2.43845137e-06				

III VALIDATION FRAMEWORK

The validation of the L3 wind products encompasses the NRT and the MY validation.

The quality of the High-Resolution Wind SAR L3 products relies heavily on the quality of the L2 products (referred to as L2-HRWIND) generated by the HR-WIND L2-to-L2 processing chain, which performs a dual-pol wind inversion.

The quality of the L2-HRWIND products is controlled through comparisons with:

1. In-situ data: buoys from CMEMS (for NRT)
2. In-situ data: buoys from CMEMS (for NRT) NWP (Numerical Weather Prediction) models: ECMWF IFS, both for MY and NRT and NCEP for NRT only
3. Remote sensing data: HY2 scatterometer (for MY)

For the NRT stream, the validation also includes:

- high level monitoring of the daily L3 products (such as number (Nb) of available and valid pixels) and
- a joint-validation of S1A and S1C L3 products against NWP models (both NCEP and ECMWF IFS) to control the quality of the newly included S1C data.

III.1 Special note on IPF versions

The High-Resolution Wind SAR L3 products are based on upstream ESA data processed with different versions of the Instrument Processing Facility (IPF). The current version is 3.90 (for data processed after 2024-11-25 15:17:24). Ultimately, the MY dataset will cover around ten IPF versions, starting with IPF version 2.90. Extending the dataset further into the past would require a full reprocessing of the input database by ESA, as versions earlier than 2.90 are not compatible with the current dual-pol wind processing. The limitation related to IPF version before 2.90 is due to the fact that the co- and cross-polarization NRCS variables, main input for the wind inversion, were not included for earlier IPF versions. Therefore, for the MY comparisons with NWP models, metrics are also reported by IPF version.

III.2 Special note on dual-pol algorithm and wind regimes

The dual-pol algorithm has been developed and validated for high-speed wind conditions (i.e., Tropical Cyclone studies). A summary of this validation is given in the Annex to this document. In European seas, the winds are usually more moderate, therefore the validity of the dual-pol is still under careful investigation. The CMEMS L3 products are expected to support further evaluation of this algorithm for moderate winds in future versions. Thus, all comparisons are also performed with the results of the first step of the algorithm (co-pol only) to stress the difference between the co-pol and dual-pol methods.

III.3 Comparisons with in-situ measurements

The reference dataset for in-situ measurement is the dataset of moored buoys from the CMEMS In-Situ TAC. As the SAR derived wind is equivalent to the neutral wind at 10 m above the sea level, a vertical profile model is used to extrapolate the buoy-measured wind speed to 10 m altitude.

$$U_{10m} = U_z \frac{10^{0.11}}{z} \tag{7}$$

Some buoys from the CMEMS In-situ TAC were excluded from the validation for the following reasons:

(i) buoys known to show abnormally high winds such as ASTAN, POEM and Carnot,

(ii) buoys for which no altitude is known (preventing the vertical extrapolation),

(iii) buoys with altitudes higher than 8 m, often corresponding to wide offshore platforms. Such structures may generate bright target on SAR images, even when an *a priori* bright-target detection is applied.

III.4 Comparisons with NWP models

III.4.1 L2 comparisons

In addition to the comparisons performed against the in-situ measurements, NWP models are also used to validate the internal L2 HRWIND products. For NRT, the comparison is performed over a 12-days cycle (minimum period of time for each location to be sampled) during March 2024 against two NWP models for co-located products over Europe: NCEP and ECMWF IFS. For MY, the comparison is performed against ECMWF IFS only, but over a longer period (extract of a 2000 L2 S1A slice per region from 2021-2023 and 500 L2 S1B slices per regions from 2018-2021).

Both models have a coarser resolution (~10°) than the L2 HRWIND products. Therefore, for comparison purposes, the HR-WIND products were degraded to match the NWP model resolution, i.e., the SAR images were first smoothed to the NWP resolution and then downsampled onto the model grid. The comparison is performed for data with a “good” quality level.

III.4.2 L3 comparisons

In order to evaluate the quality of the newly included S1C data, the performances of L3 S1C products against NWP models are compared with the performances of L3 S1A products against the same NWP models. These L3 comparisons are performed over a 12-days cycle taken during August 2025 (when both S1A and S1C are available), against both NCEP and ECMWF IFS.

To account for the coarser resolution of the NWP models, the L3 products were averaged over tiles (of 10 by 10 pixels) only keeping those where more than 50% of the pixels averaged have the best quality level (‘quality_level==4’).

III.5 Comparisons with in-situ measurements

The reference dataset for in-situ measurements is the dataset of moored buoys provided by the CMEMS In-Situ TAC. Since SAR-derived winds are equivalent to the neutral wind at 10 m above the sea level, a vertical profile model is used to extrapolate the wind measured by the buoy-measured wind speed to a 10 m altitude.

$$U_{10m} = U_z \frac{10^{0.11}}{z} \tag{7}$$

Some buoys from the CMEMS In-situ TAC were excluded from the validation for the following reasons::

(i) those known to show abnormally high winds such as ASTAN, POEM and Carnot,

(ii) those for which no altitude is known (preventing the vertical extrapolation),

(iii) those with an altitude higher than 8 m, often corresponding to large offshore platforms. Such structures may generate bright target on SAR images, even when an *a priori* bright-target detection is applied.

III.6 Comparisons with scatterometer L2 products

The comparison with scatterometer L2 products are only available for MY production stream. The scatterometer products considered here are derived from HY-2 and processed by OSI SAF.

Two datasets are considered co-located when the time difference between acquisitions is less than 120 minutes and their common spatial coverage exceeds 30 km^2.

III.7 Metrics considered

For each comparison, three metrics are computed:

- the mean difference (i.e. bias) between SAR products (both L2 and L3) and the reference dataset;
- the Root-Mean-Square Deviation (RMSD) between SAR products and the reference dataset;
- the Scatter Index (SI) between SAR products and the reference dataset.

For visual comparison, scatter plots are shown as histograms and may be displayed along with the Q-Q plots , which compare the quantiles of the two probablility distributions.

¡Error! No se encuentra el origen de la referencia. Table 10 summarizes the datasets used for these valid ations, along with the associated metrics.

The metrics may be computed over the entire European Seas, or over one of the five CMEMS regions: Arctic Ocean (ARC), Norther Atlantic Ocean (ATL), Baltic Sea (BAL), Black Sea (BLK) or Mediterranean Sea (MED).

Table 10 : Summary of reference datasets considered for validation along with the metrics associated and the document where to find these definitions. <type> stands for all four metrics: 'VISUAL' (=scatterplot), 'BIAS', 'RMSD' and 'SI'. The <geog> stands for 4 CMEMS regions ('ARC','ATL','BAL','MED'), when the <geog> tag is not present, the metrics are computed over the entire CMEMS European seas. The 'BLK' region has not been included yet as the number of available colocations was considered too low.

Stream	Product Level	Reference datasets	Time period	Associated metrics (*and document*)
NRT	L2-HRWIND	L3 NRT product	18/06/2024 to 25/09/2024	WIND-D-NC-AVAIL-<geog> WIND-D-NC-VALID-<geog> (*ScVP*)
NRT	L2-HRWIND	CMEMS moored buoys (INSITU_GLO_NRT_OBSERVATIONS_013_030)	2021- 2023 (subset)	WIND_spd-CLASS2-MOOR-<type> (*ScVP*)
NRT	L2-HRWIND	ECMWF IFS, NCEP	01/03/23 to 12/03/23	WIND_spd-CLASS4-MOD_emcwf_ifs-<type>-12D WIND_spd-CLASS4-MOD_ncep-<type>-12D (*ScVP*)
NRT	L3	ECMWF IFS, NCEP	01/08/25 to 12/08/25	WIND_spd-CLASS4-MOD_emcwf_ifs-<type>-12D WIND_spd-CLASS4-MOD_ncep-<type>-12D (*ScVP*)
MY	L2-HRWIND	HY2 scatterometer L2 product	2019- 2023 (subset)	WIND_spd-CLASS4-L2SCAT_hy2-<type>-<geog> (*ScMYVP*)
MY	L2-HRWIND	ECMWF IFS	2018- 2023 (subset)	WIND_spd-CLASS4-MOD_emcwf_ifs-<type>-<geog> WIND_spd-CLASS4-MOD_emcwf_ifs-<type> WIND_dir-CLASS4-MOD_emcwf_ifs-<type> (*ScMYVP*)

IV VALIDATION RESULTS

IV.1 NRT Validation

This section provides the results of the validation for the Near Real Time production stream. It contains PQ information about the availability and validity of data per day and per dataset, and results of comparison with in-situ measurements and with NWP models.

IV.1.1 Internal validation: WIND-D-NC-AVAIL & WIND-D-NC-VALID

In this section, we present the Product Quality (PQ) information about the availability and validity of NRT data per day and dataset, available since the start of the NRT product in June 2024. Here, one dataset corresponds to a specific region and one orbit orientation (ascending – ASC – or descending – DSC).

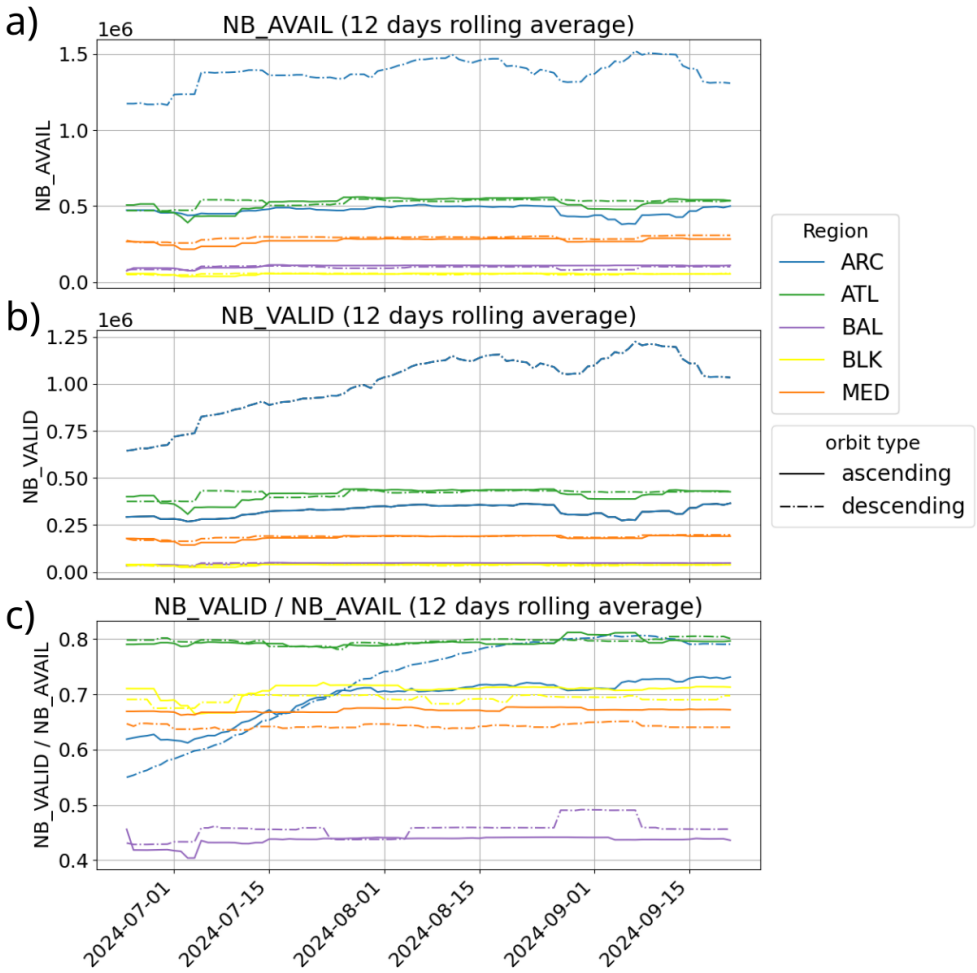

Figure 2: Number of pixels available (a) and valid (b) per day, smoothed over 12 days to account for the satellite orbit. (c) Ratio valid/available.

Figure 2 shows that the statistics of availability and validity of pixels have remained relatively stable since the EIS June 2024 and the start of the NRT production. For the Arctic (ARC) zone, shown in blue in the panels of **¡Error! No se encuentra el origen de la referencia.**, the ratio of valid pixels is increasing from June to September, which may be related to the melting ice in the Arctic Ocean.

Overall, closed seas have a smaller valid ratio compared to the open ocean. In particular, the Baltic Sea (BAL) is the zone for which the ratio of valid data over available data is the smallest (around 44%). This could be explained by the complexity of this semi-enclosed sea that may trigger various flags, such as the land flag, the ice flag, the bright target flag, among others.

IV.1.2 Comparisons with in-situ measurements

In this section, we present the results of our comparison between HR-WIND L3 NRT products and in-situ wind measurements obtained from moored buoys provided by the In-Situ TAC.

¡Error! No se encuentra el origen de la referencia..a shows the selection of 232 co-localisations for comparisons between buoys and Sentinel-1A data. The results of these co-localisations (WIND_spd-CLASS2-MOOR-VISUAL) are shown in Figure 3.b and Table 11. The bias and Scatter Index (SI) for the co-pol algorithm are respectively -1.84 m/s and 21.85%. These metrics are slightly better for the dual-pol algorithm (–1.67 m/s and 21.48%, respectively). The Q-Q , which compares the quantiles of the two datasets, indicates that the dual-pol algorithm is slightly less biased than the co-pol algorithm across all wind ranges, including moderate winds. This suggests the relevance of the dual-pol wind inversion in all wind conditions; however, the limited number of co-localisations prevents drawing definitive conclusions.

Table 11 : Metrics for In-situ comparison of co-localisations between buoys and Sentinel-1A data.

Metric	Dual-pol algorithm	Co-pol algorithm
WIND_spd-CLASS2-MOOR-BIAS	-1.67 m/s	-1.84 m/s
WIND_spd-CLASS2-MOOR-RMSD	2.69 m/s	2.74 m/s
WIND_spd-CLASS2-MOOR-SI	21.48 %	21.85 %

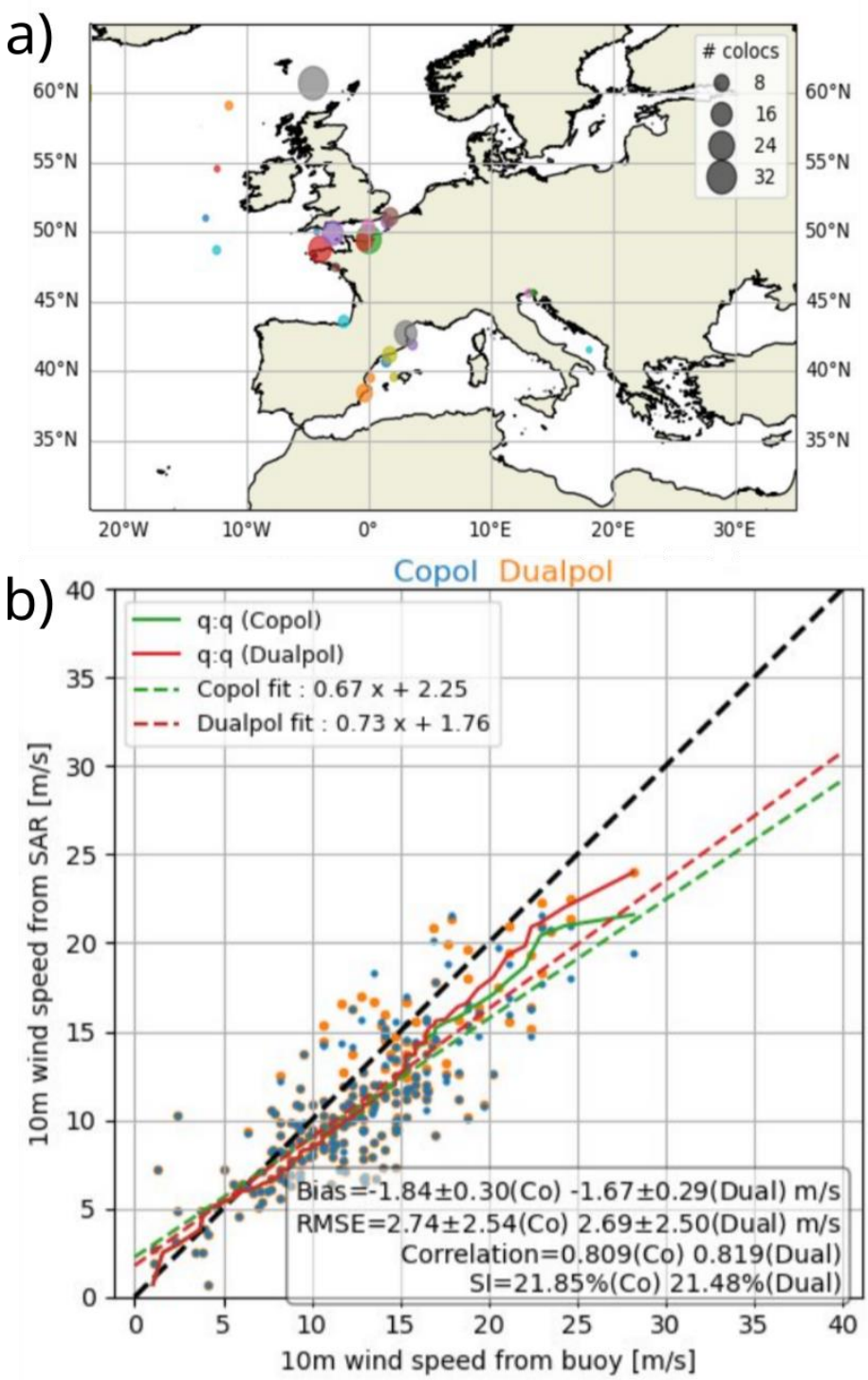

Figure 3: Position of buoys co-located with SAR data (a). The size of the marker represents the number of co-localisations per buoy. (b) 10 m wind speed from SAR data for the co-pol (in blue) and the dual-pol (in orange) compared with the 10 m wind speed obtained from co-located buoys.

IV.1.3 Comparisons with NWP models

IV.1.3.1 L2 comparisons

This subsection presents the results of the comparison between L2 internal products and NWP models..

Figure 4 shows the spatial distribution of the HR-WIND products considered in the analysis and the co-located NWP model outputs available at the time of the assessment. For the models sourced by ECMWF, there is a model file for each SAR-derived wind measurements.

Figure 5 shows the scatterplots of the L2 product with dual-pol (i.e. L2-HRWIND product) against the 2 NWP models both for the IW DV and EW DH acquisition and polarization modes (i.e., the dominant modes). The associated global bias and RMSD are shown in Table 12. The poorer performance observed for the EW-DH mode—characterized by a stronger bias and larger RMSD compared to IW-DV—is attributed to the less favorable DH polarization and to potential inaccuracies in sea-ice masking within the SAR imagery, which may lead to overestimated wind speeds.

A comparison of the dual-pol (red line) and co-pol (magenta line) Q-Q plots in Figure 5, shows similar behavior for wind speeds below 15 m/s. Above 15 m/s the dual-pol algorithm exhibits larger deviation from the NWP models. This effect was expected, as a consequence of the known underestimation of high winds by NWP models.

Table 13 shows more comprehensive statistics, including both IW-DH and EW-DV acquisition and polarisation modes. It also includes the biases and RMSD associated with the co-pol wind inversion method. The statistics are computed for three different wind ranges – [0; 5], [5; 15], [15; inf] – to highlight differences in behaviour. The first range (below 5 m/s) corresponds to low wind speeds for which the cross-polarization NRCS is dominated by noise, thus the dual-pol output is forced to be the same as for the co-pol. The 5-15 m/s range corresponds to the most frequent winds (equivalent to quantiles 20 to 90, with a peak occurrence around 7.5 m/s). Above 15 m/s (10% of higher speed winds), the impact of the dual-pol compared to the co-pol becomes significant.

Table 12 : Global performance -DH acquisition (Acq.) and polarization (Pol.) modes

Acq. Mode Pol. mode	NWP model	Number of L2 files	Bias (m/s) WIND_spd-CLASS4-<MOD>-BIAS-12D	RMSD (m/s) WIND_spd-CLASS4-<MOD>-RMSD-12D
IW - DV	NCEP	213	0.18	1.85
	ECMWF	662	0.30	1.64
EW - DH	NCEP	36	0.74	2.19
	ECMWF	146	1.56	3.11

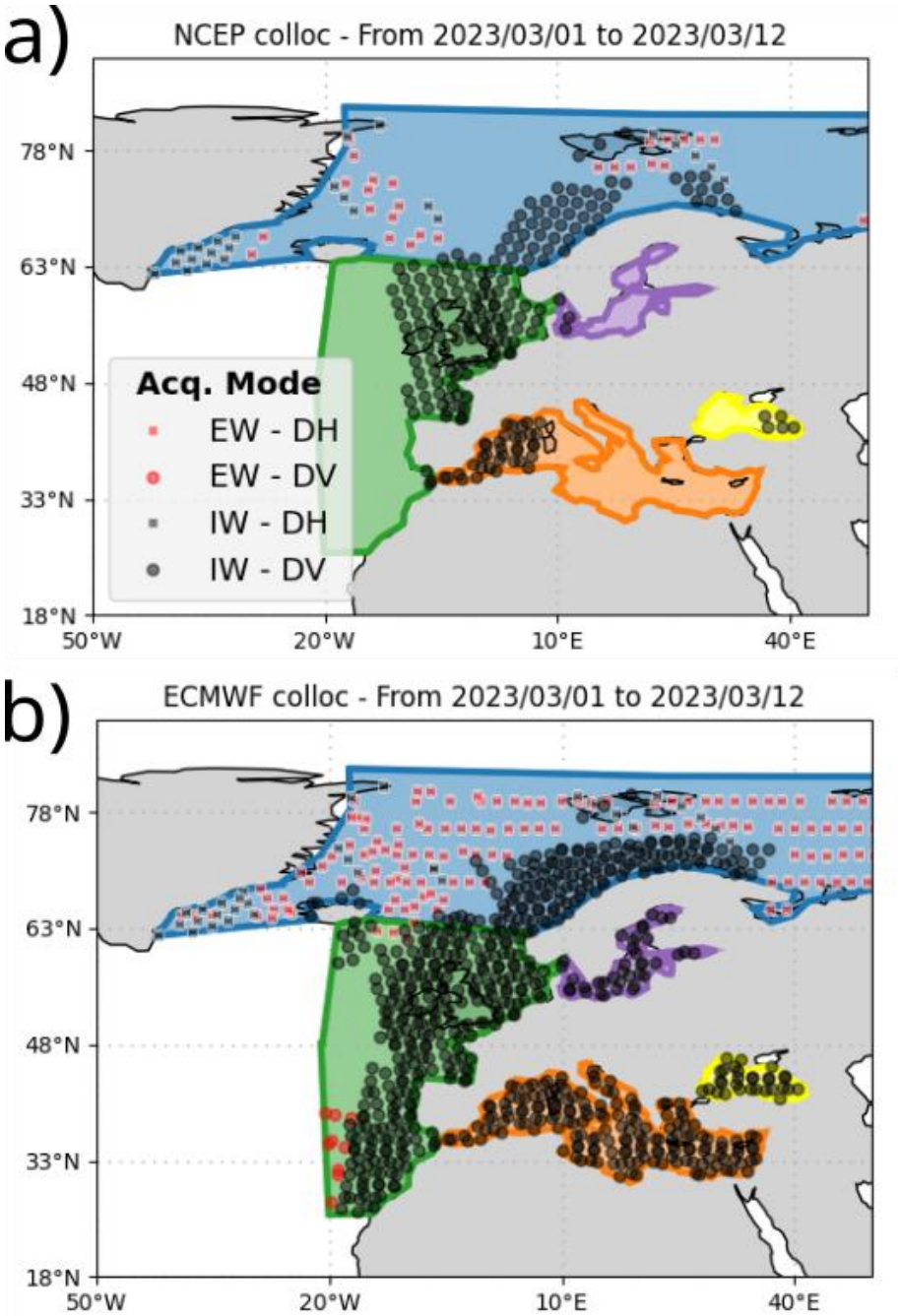

Figure 4: Position of the L2 HRWIND products co-located with a) NCEP winds and b) ECMWF winds for one S1A cycle from 01/03/2023 to 12/03/2023 over the CMEMS Regions. Each region is represented by a coloured polygon. The red squares represent measurements performed with the EW-DH acquisition configuration, the red circles stand for EW-DV acquisitions, the black squares represent IW-DH acquisitions, and the black circles represent IW-DV acquisitions.

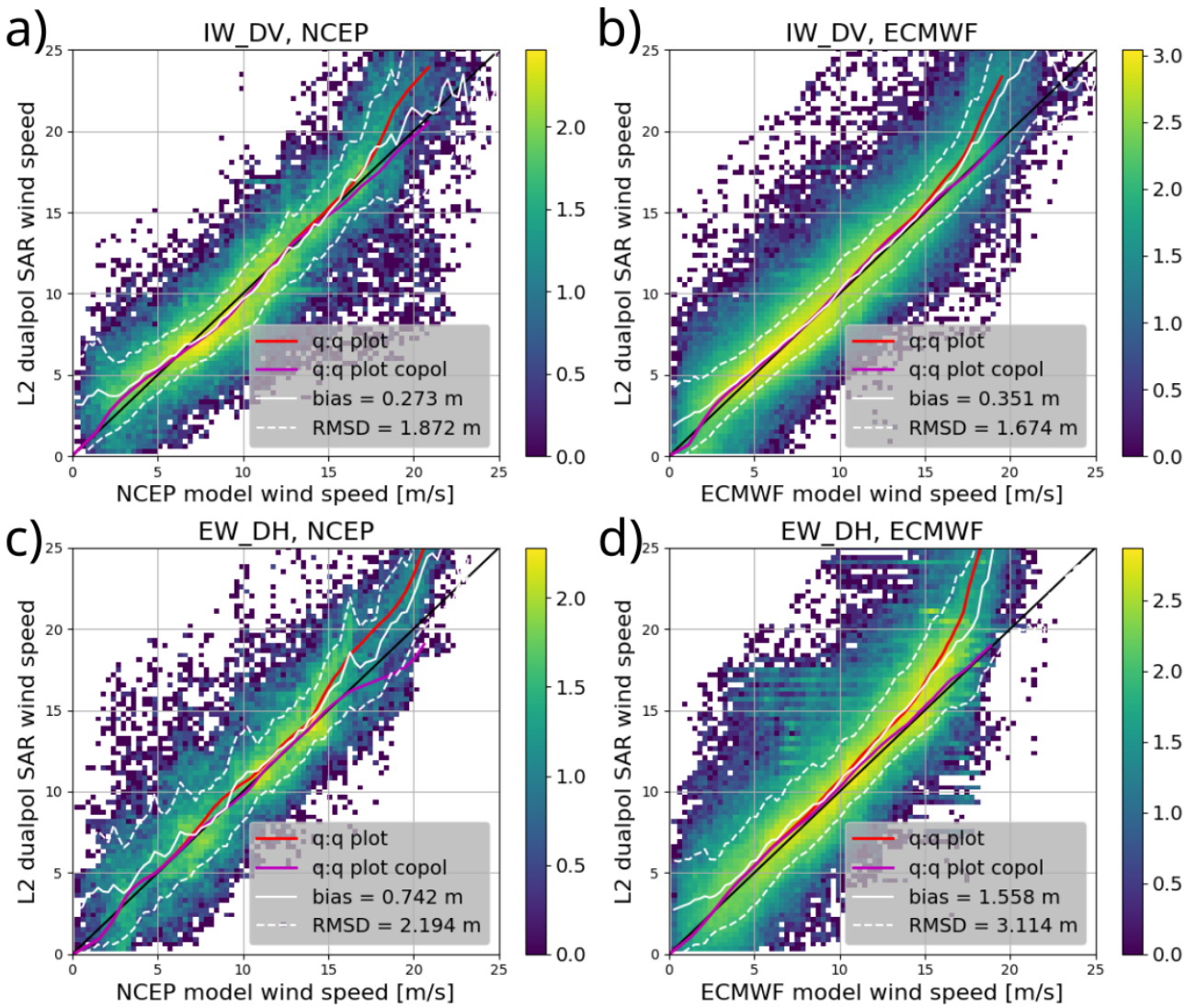

Figure 5: WIND_spd-CLASS4-<MOD>-VISUAL-12D: Scatterplots (shown as histograms) for the L2-HRWIND products against NCEP winds – a) and c) – and ECMWF winds – b) and d). Panels a) and b) correspond to the IW – DV acquisition and polarisation mode, whereas panels c) and d) show results for the EW-DH mode. The red line represents the Q-Q plot of the data shown. For comparison, the Q-Q plot of a co-pol wind inversion is shown in magenta

Table 13a : Performance of L2 products from dual-pol (HR WIND) and co-pol (IPF) wind inversion schemes against various NWP models. Results are provided by wind range, for different acquisition (Acq.) and polarization (Pol.) mode. U < 5 m/s

Acq. Mode + Acq. Pol	Wind model	Number of obs	U < 5 m/s			
			Mean Bias		RMSD	
			dual-pol	co-pol	dual-pol	co-pol
IW DV	NCEP	213	0.04	0.04	1.40	1.40
	ECMWF	662	-0.05	-0.05	1.16	1.16
IW DH	NCEP	24	-0.75	-0.75	1.28	1.28
	ECMWF	32	-0.79	-0.79	1.10	1.10
EW DV	NCEP	0	-	-	-	-
	ECMWF	9	-0.12	-0.12	0.71	0.71
EW DH	NCEP	36	-0.61	-0.62	1.43	1.45
	ECMWF	146	-0.85	-0.79	1.94	1.94

Table 13b : Performance of L2 products from dual-pol (HR WIND) and co-pol (IPF) wind inversion schemes against various NWP models. Results are provided by wind range, for different acquisition (Acq.) and polarization (Pol.) modes. 5 m/s < U < 15 m/s

Acq. Mode + Acq. Pol	Wind model	Number of obs	5 m/s < U < 15 m/s			
			Mean Bias		RMSD	
			dual-pol	co-pol	dual-pol	co-pol
IW DV	NCEP	213	-0.12	-0.13	1.59	1.65
	ECMWF	662	0.15	0.14	1.44	1.45
IW DH	NCEP	24	0.83	0.83	2.10	2.10
	ECMWF	32	0.34	0.34	1.25	1.26
EW DV	NCEP	0	-	-	-	-
	ECMWF	9	-1.23	-1.24	1.49	1.50
EW DH	NCEP	36	0.04	-0.35	1.57	1.93
	ECMWF	146	0.29	0.02	1.74	1.77

Table 13c : Performance of L2 products from dual-pol (HR WIND) and co-pol (IPF) wind inversion schemes against various NWP models. Results are provided by wind range, for different acquisition (Acq.) and polarization (Pol.) modes. U > 15 m/s

Acq. Mode + Acq. Pol	Wind model	Number of obs	U > 15 m/s			
			Mean Bias		RMSD	
			dual-pol	co-pol	dual-pol	co-pol
IW DV	NCEP	213	1.99	0.13	3.03	1.31
	ECMWF	662	2.23	0.20	3.23	1.08
IW DH	NCEP	24	2.59	1.77	3.22	2.35
	ECMWF	32	1.26	0.68	1.69	1.02
EW DV	NCEP	0	-	-	-	-
	ECMWF	9	-	-	-	-
EW DH	NCEP	36	3.04	-1.49	3.63	3.22
	ECMWF	146	3.69	0.49	4.67	2.43

IV.1.3.2 L3 comparisons

This subsection presents the results of the L3 comparison with the NWP models.

The scatterplots of the L3 SAR Wind products from S1A and S1C against the 2 NWP models are shown in Figure 6 for the IW DV acquisition and polarization mode and in Figure 7 for the EW DH acquisition and polarization modes (i.e., the dominant modes). The global bias, RMSD and SI associated are shown in Table 14. Bias is slightly positive in IW DV, but negative and larger in magnitude for EW DH. However, RMSD and SI are equivalent for all main modes and satellites, except for IW DV S1A, for which the wind inversion was originally calibrated.

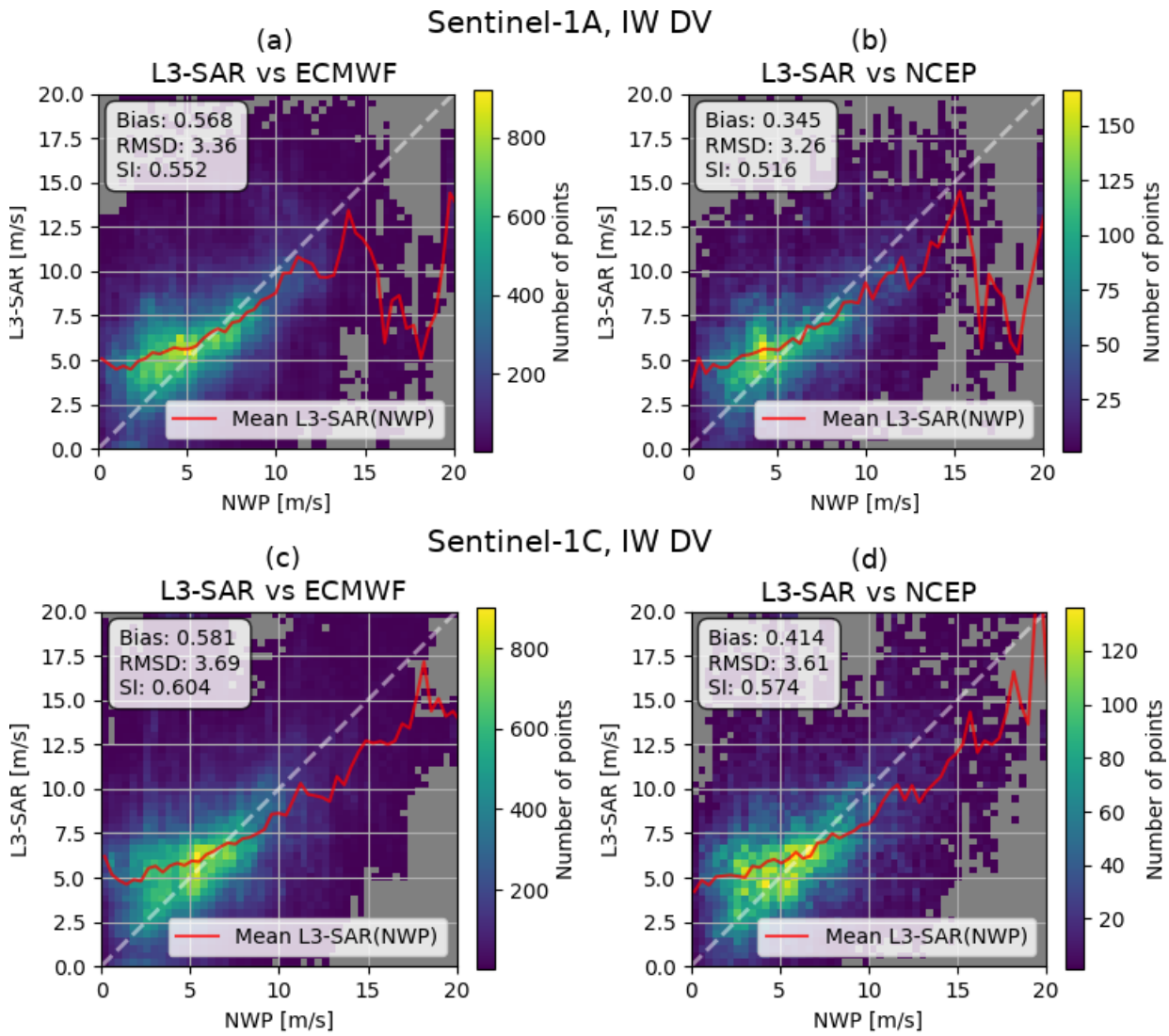

*Figure 6: WIND_spd-CLASS4-<MOD>-VISUAL-12D: Scatterplots (shown as histograms) for the L3 HR-WIND products against ECMWF winds – a) and c) – and NCEP winds – b) and d). Panels a) and b) correspond to S1A while panels c) and d) correspond to S1C. The figure shows the results for the **IW – DV** acquisition and polarisation mode. The red line represents the average value of L3 HR-WIND wind speed per NWP wind speed bin.*

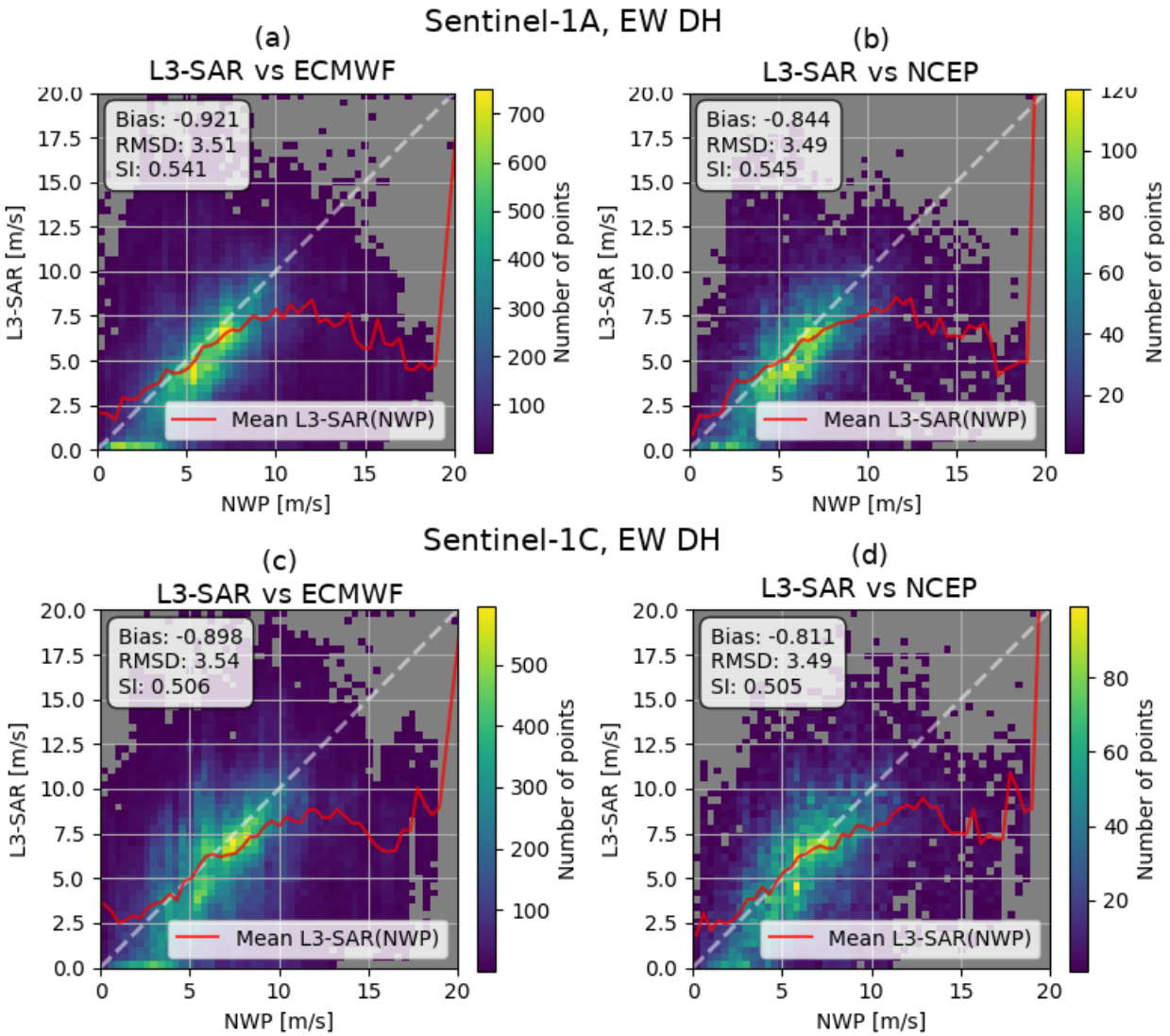

*Figure 7: WIND_spd-CLASS4-<MOD>-VISUAL-12D: Scatterplots (shown as histograms) for the L3 HR-WIND products against ECMWF winds – a) and c) – and NCEP winds – b) and d). Panels a) and b) correspond to S1A while panels c) and d) correspond to S1C. The figure shows the results for the **EW – DH** acquisition and polarisation mode. The red line represents the average value of L3 HR-WIND wind speed per NWP wind speed bin.*

Table 15 shows more comprehensive statistics, including both the IW-DH and EW-DV acquisition and polarisation modes. The statistics are computed for three different wind ranges – [0; 5], [5; 15], [15; inf] – to highlight the different behaviours.

We stress that all acquisition-polarization combinations do not represent an equivalent number of points. IW DH and EW DV are scarce w.r.t the number of points for IW DV and EW DH. In addition, high wind speeds are also under-represented due to the temporal distribution of the comparison points between Sentinel-1A and Sentinel-1C, which led to the selection of summer conditions (first cycle of August).

Comparisons between Sentinel-1A and Sentinel-1C at intermediate wind speed show similar statistics, with minor differences attributed to statistical variation of the comparison points. A low wind speed in IW DV, Sentinel-1C displays a lower bias than Sentinel-1A, which is expected because of the lower SNR.

However, longer time series will be required to draw conclusive results that eliminate seasonal variations.

Table 14 : Global performance of the HR-WIND products (dual-pol method) against NWP models, for IW-DV and EW-DH acquisition (Acq.) and polarization (Pol.) modes

Acq Mode Pol. Mode	Satellite	NWP model	Bias (m/s) WIND_spd-CLASS4-<MOD>-BIAS-12D	RMSD (m/s) WIND_spd-CLASS4-<MOD>-RMSD-12D	SI (%) WIND_spd-CLASS4-<MOD>-SI-12D
IW - DV	S1A	NCEP	0.36	3.24	51
		ECMWF	0.58	3.33	55
EW - DH	S1A	NCEP	-0.83	3.49	55
		ECMWF	-0.91	3.51	54
IW - DV	S1C	NCEP	0.41	3.52	56
		ECMWF	0.58	3.58	58
EW - DH	S1C	NCEP	-0.83	3.49	51
		ECMWF	-0.91	3.55	51
All	S1A	NCEP	0.05	3.30	52
		ECMWF	0.20	3.38	55
All	S1C	NCEP	-0.05	3.51	54
		ECMWF	0.03	3.57	56
IW - DV	S1A & S1C	NCEP	0.38	3.34	53
		ECMWF	0.58	3.42	56
EW - DH	S1A & S1C	NCEP	-0.83	3.49	53
		ECMWF	-0.91	3.53	52

Table 15a : Performance of L3 SAR Wind products against various NWP models. Results are provided by wind range, for different acquisition (Acq.) and polarization (Pol.) modes and for both S1A and S1C. U < 5 m/s

SAT	Acq. Mode + Acq. Pol	Wind model	U < 5 m/s		
			Mean Bias (m)	RMSD (m)	SI
S1A	*IW DV*	NCEP	0.23	1.58	0.54
		ECMWF	0.33	1.68	0.61
	IW DH	NCEP	0.03	1.87	0.60
		ECMWF	0.10	1.82	0.61
	EW DV	NCEP	1.94	2.45	1.03
		ECMWF	1.60	2.28	0.86
	EW DH	NCEP	-0.91	1.70	0.53
		ECMWF	-0.99	1.79	0.56
S1C	*IW DV*	NCEP	0.08	1.59	0.54
		ECMWF	0.10	1.67	0.59
	IW DH	NCEP	-0.97	1.61	0.43
		ECMWF	-0.97	1.74	0.49
	EW DV	NCEP	1.20	2.12	0.99
		ECMWF	1.19	2.19	1.01
	EW DH	NCEP	-0.90	1.78	0.55
		ECMWF	-0.98	1.90	0.60

Table 15b : Performance of L3 SAR Wind products against various NWP models. Results are provided by wind range, for different acquisition (Acq.) and polarization (Pol.) modes and for both S1A and S1C. 5 m/s < U < 15 m/s

SAT	Acq. Mode + Acq. Pol	Wind model	5 m/s < U < 15 m/s		
			Mean Bias (m)	RMSD (m)	SI
S1A	*IW DV*	NCEP	-0.23	2.44	0.28
		ECMWF	-0.08	2.39	0.28
	IW DH	NCEP	0.18	2.33	0.33
		ECMWF	0.24	2.32	0.33
	EW DV	NCEP	1.88	2.93	0.45
		ECMWF	1.39	2.62	0.39
	EW DH	NCEP	-0.36	2.45	0.30
		ECMWF	-0.32	2.42	0.30
S1C	*IW DV*	NCEP	-0.30	2.54	0.30
		ECMWF	-0.10	2.49	0.30
	IW DH	NCEP	1.68	3.68	0.49
		ECMWF	1.71	3.76	0.49
	EW DV	NCEP	-0.11	1.66	0.24
		ECMWF	-0.15	1.80	0.26
	EW DH	NCEP	-0.07	2.64	0.32
		ECMWF	-0.11	2.58	0.31

Table 15c : Performance of L3 SAR Wind products against various NWP models. Results are provided by wind range, for different acquisition (Acq.) and polarization (Pol.) modes and for both S1A and S1C. U > 15 m/s

SAT	Acq. Mode + Acq. Pol	Wind model	U > 15 m/s		
			Mean Bias (m)	RMSD (m)	SI
S1A	*IW DV*	NCEP	-0.94	2.71	0.15
		ECMWF	-1.88	3.26	0.18
	IW DH	NCEP			
		ECMWF			
	EW DV	NCEP			
		ECMWF			
	EW DH	NCEP	0.82	0.82	0.04
		ECMWF	1.52	2.85	0.17
S1C	*IW DV*	NCEP	1.17	1.99	0.12
		ECMWF	1.13	2.00	0.12
	IW DH	NCEP			
		ECMWF			
	EW DV	NCEP			
		ECMWF			
	EW DH	NCEP	-0.82	1.83	0.10
		ECMWF	3.04	3.54	0.22

IV.2 MY Validation

This section provides the results of the validation for the Multi Year production stream. It contains results of comparison with NWP models and with HY2 scatterometers.

IV.2.1 Comparison with NWP model

This section presents the results of the comparison between SAR MY L3 products and collocated ECMWF IFS NWP model, for both for wind speed and wind direction. This section also provides an estimate of the evolution of the bias and RMSD with the evolution of the upstream processing.

Figure 8 shows WIND_spd-CLASS4-MOD_emcwf_ifs-VISUAL-<geog>, i.e. the scatterplot of SAR L2 wind speed against ECMWF IFS model values by region (each subplot represents one CMEMS region), for S1A only. The associated metrics, WIND_spd-CLASS4-MOD_emcwf_ifs-<type>-<geog> metrics (for <type> = 'BIAS', 'RMSD' or 'SI') are reported in Table 16 for both S1A and S1B for the dual-pol algorithm along with the co-pol algorithm metrics.

Table 17 provides the WIND_dir-CLASS4-MOD_emcwf_ifs-<type>-<geog> metrics (for <type> = 'BIAS', 'RMSD' or 'SI') for both S1A and S1B.

Table 16: Wind speed metrics (bias, Root-Mean-Square Difference -RMSD- and Scatter Index -SI-) for comparison between HR-WIND MY S1A datasets and ECMWF IFS NWP model, for the "dual-pol" and "co-pol" algorithms. ALL: all regions; ARC: Arctic; ATL: Atlantic; BAL: Baltic; BLK: Black Sea; MED: Mediterranean

Satellite	WIND_spd	Metric	ALL	ARC	ATL	BAL	BLK	MED
S1A	dual-pol	BIAS [m/s]	-0.37	-0.74	-0.03	-0.88	0.02	-0.23
		RMSD [m/s]	4.34	4.53	2.00	8.09	1.83	2.13
		SI [%]	45.97	42.89	23.32	67.58	29.07	30.53
	co-pol	BIAS [m/s]	-0.02	-0.20	0.16	0.25	0.09	-0.09
		RMSD [m/s]	2.16	2.38	1.43	2.04	1.63	1.54
		SI [%]	26.20	25.91	17.41	25.15	26.50	23.40
S1B	dual-pol	BIAS [m/s]	0.08	-0.01	0.18	-0.25	-0.22	0.01
		RMSD [m/s]	3.00	3.17	1.99	6.48	2.17	2.08
		SI [%]	33.12	32.94	23.31	65.91	32.33	30.12
	co-pol	BIAS [m/s]	0.18	0.15	0.28	0.41	-0.12	0.09
		RMSD [m/s]	1.82	1.81	1.47	2.16	1.46	1.59
		SI [%]	21.00	19.88	17.47	29.12	22.76	23.73

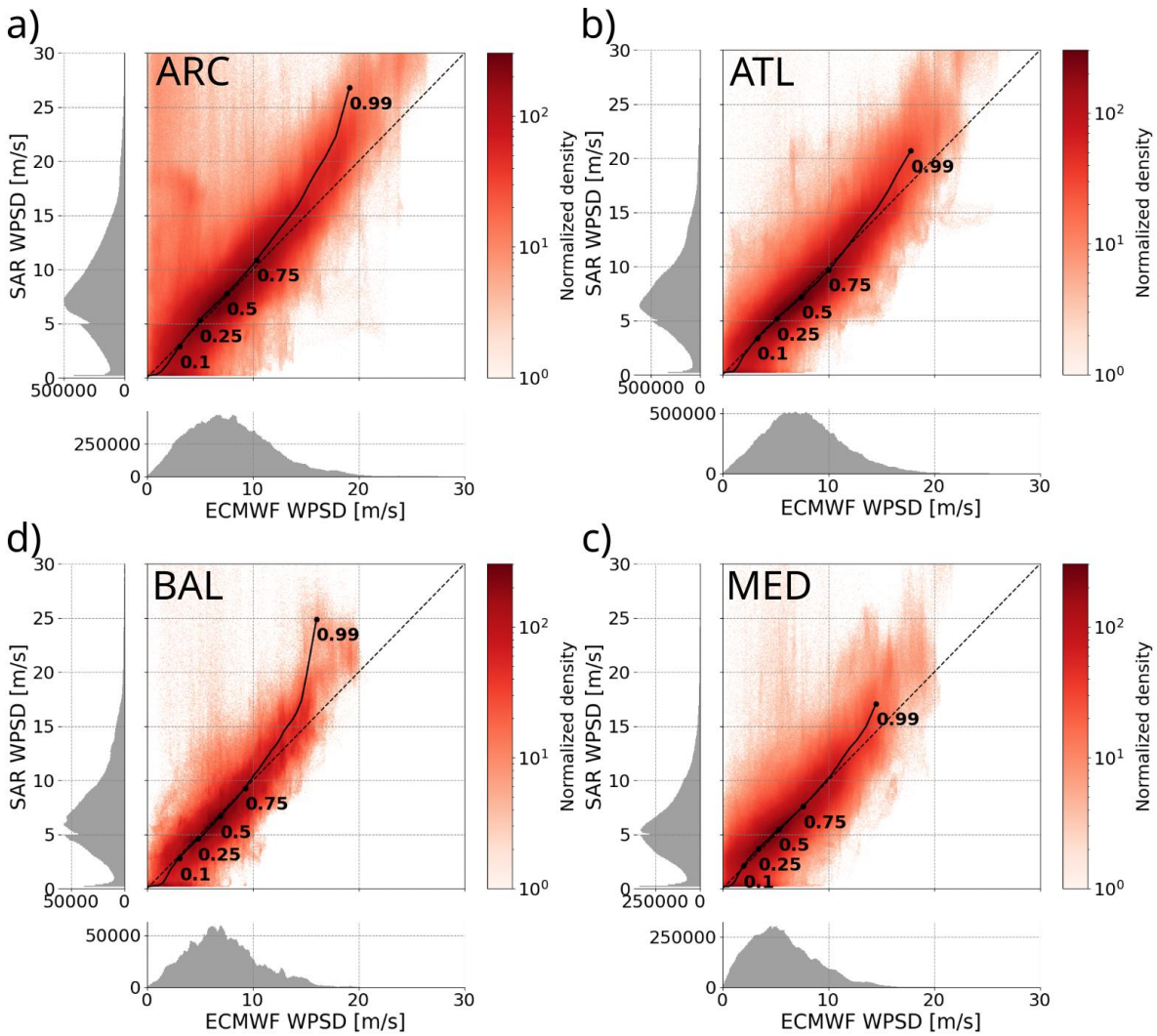

Figure 8: WIND_spd-CLASS4-MOD_ecmwf_ifs-VISUAL-<geog>: Scatterplots (shown as histograms) for the HR-WIND MY products (S1A only) against ECMWF winds for four CMEMS region: ARC (Arctic), ATL (Atlantic), BAL (Baltic) and MED (Mediterranean). The 1D histograms of SAR and model wind speed are also depicted in grey

Table 17: Wind direction metrics for comparison between HR-WIND MY products and ECMWF IFS NWP model. WIND_dir-CLASS4-MOD_emcwf_ifs-<type>-<geog> for the "dual-pol" and "co-pol" algorithms. ALL: all regions; ARC: Arctic; ATL: Atlantic; BAL: Baltic; BLK: Black Sea; MED: Mediterranean

Satellite	WIND_dir	type[units]	ALL	ARC	ATL	BAL	BLK	MED
S1A	dual-pol & co-pol	BIAS [°]	-0.16	0.60	0.27	1.79	-2.44	0.64
		RMSD [°]	60.75	61.03	53.52	68.08	69.13	62.10
		SI [%]	29.70	30.24	28.96	37.32	33.11	33.86
S1B	dual-pol & co-pol	BIAS [°]	0.39	1.01	0.15	-1.12	-0.70	0.94
		RMSD [°]	58.73	54.93	52.33	63.43	51.13	61.78
		SI [%]	30.37	26.81	27.25	35.94	29.49	33.79

Figure 9 shows WIND_spd-CLASS4-MOD_emcwf_ifs-VISUAL and WIND_dir-CLASS4-MOD_emcwf_ifs-VISUAL i.e., the scatterplot of SAR L2 wind speed and direction against the ECMWF IFS model for the entire CMEMS area. Figure 9.a and Figure 9.b show the differences between the dual-pol algorithm and the co-pol algorithm for S1A (over all regions). The comparison between S1A and S1B is illustrated by Figure 9.a (S1A, dual-pol) and Figure 9.c (S1B, dual-pol). Figure 9.d represents the scatterplot for the wind direction for S1A only,for both co-pol and dual-pol algorithms.

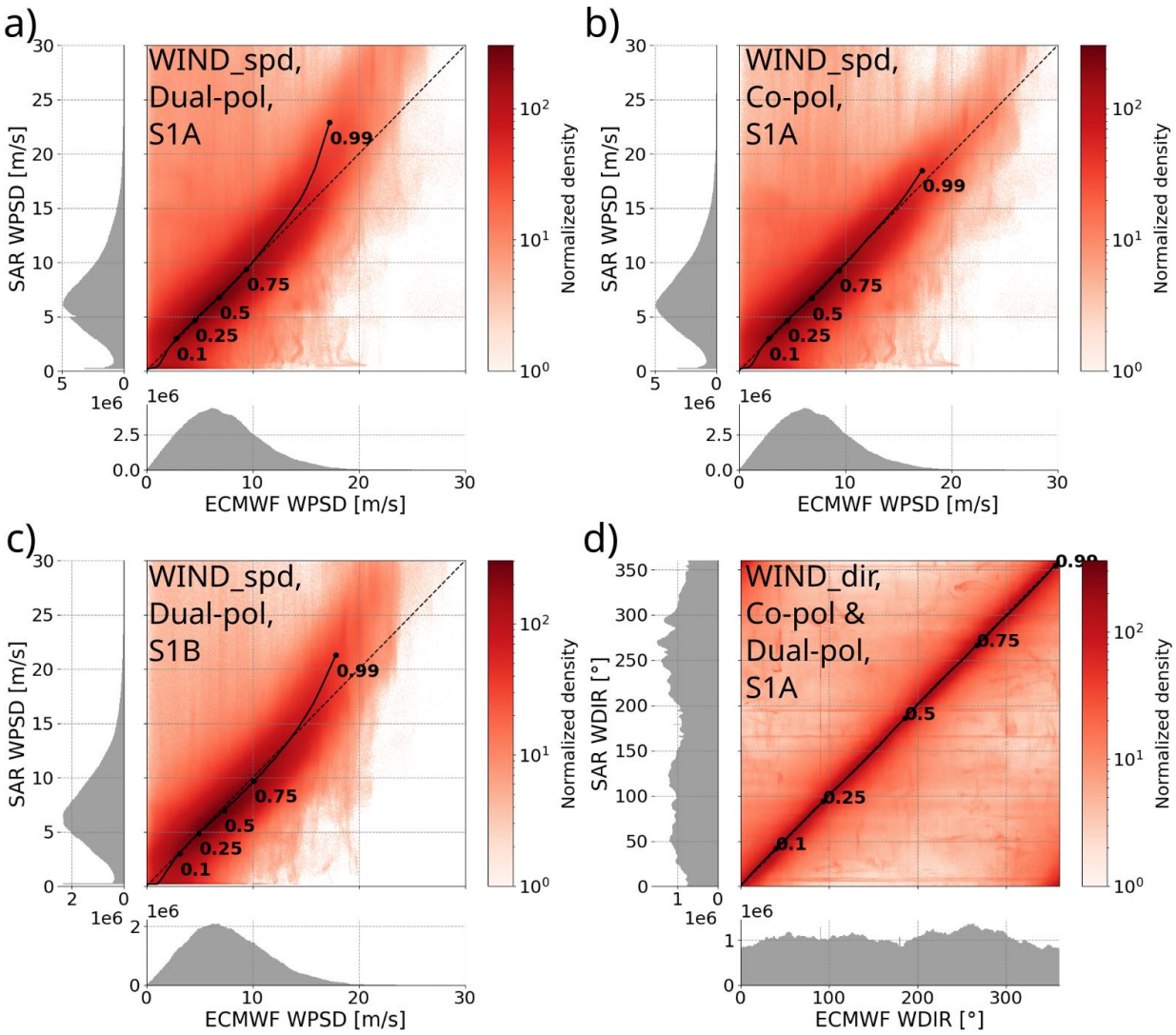

Figure 9: Scatterplots (shown as histograms) for the HR-WIND MY products against ECMWF winds, both for wind speed – a), b), c) – and wind direction – d). The scatterplots are shown both for a) and d) the S1A dual-pol algorithm, b) and d) the S1A co-pol algorithm and c)the S1B dual-pol algorithm.

As expected, the main difference between co-pol and dual-pol occurs at wind speeds above 10 to 15 m/s, where dual-pol tends to overestimate wind speeds compared to both co-pol and ECMWF values. There is no significant difference between S1A and S1B.

Figure 1Figure 10 shows the evolution of WIND_spd-CLASS4-MOD_emcwf_ifs-BIAS and WIND_spd-CLASS4-MOD_emcwf_ifs-RMSD with changes in the IPF version, for S1A only.

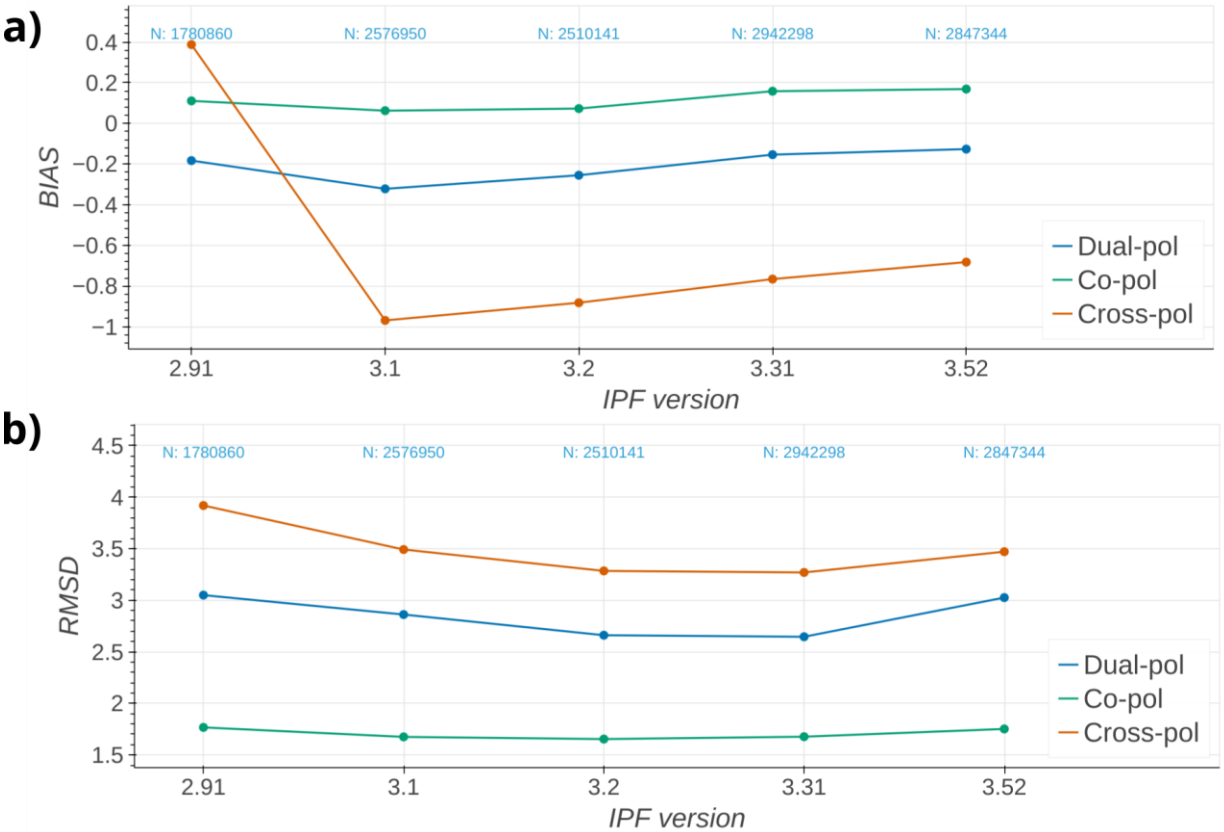

Figure 1: a) WIND_spd-CLASS4-MOD_emcwf_ifs-BIAS (top panel) and b) WIND_spd-CLASS4-MOD_emcwf_ifs-RMSD (lower panel) evolution with IPF versions for co-pol – green lines –; cross-pol – red line – and dual-pol – blue line – algorithms. Statistics performed with S1A only.

IV.2.2 Comparison with Scatterometer

This section provides comparison between SAR MY L3 products and wind measurements performed by the HY2 scatterometer. The comparison is performed for S1A only, using both the co-pol and dual-pol algorithm.

Figure 11 shows WIND_spd-CLASS4-L2SCAT_hy2-VISUAL-<geog>, i.e., the scatterplot of SAR L2 wind speed against L2 wind products from HY2 scatterometer by region. Each subplot represents one CMEMS region).

Similarly to the model, the comparison shows that above 12 m/s the dual-pol algorithm makes higher wind estimates than the scatterometer.

The associated metrics, WIND_spd-CLASS4-L2SCAT_hy2-<type>-<geog> (for <type> = 'BIAS', 'RMSD' or 'SI'), are reported in Table 18 for both co-pol and dual-pol algorithm.

Table 18: Wind speed metrics for comparison between HR-WIND MY products and L2 products from HY2 scatterometer. WIND_spd-CLASS4-L2SCAT _hy2-<type>-<geog>. Algorithms: "dual-pol" and "co-pol". ALL: all regions; ARC: Arctic; ATL: Atlantic; BAL: Baltic; MED: Mediterranean. RMSD (Root Mean Square Difference; SI (Scatter Index).

WIND_spd	<type>	ARC	ATL	BAL	MED
dual-pol	BIAS [m/s]	0.165	0.107	-0.073	0.0752
	RMSD [m/s]	1.508	1.494	1.156	1.085
	SI [%]	15.32	16.01	12.23	12.28
co-pol	BIAS [m/s]	-0.153	-0.155	-0.374	-0.093
	RMSD [m/s]	1.312	1.278	1.035	0.999
	SI [%]	13.33	13.63	10.24	11.29

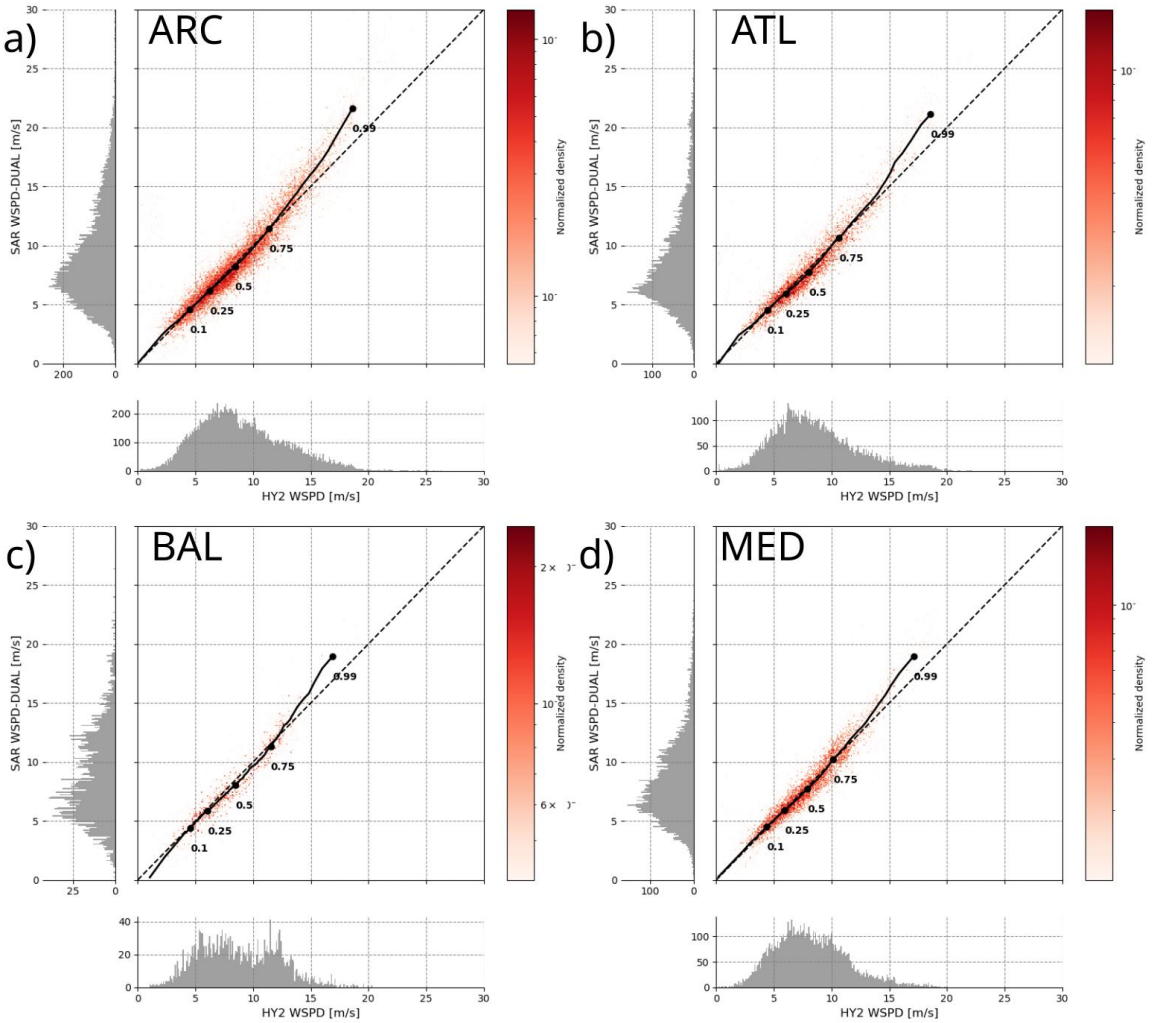

Figure 11: WIND_spd-CLASS4-L2SCAT_hy2-VISUAL-<geog>: Scatterplots (shown as histograms) for the HR-WIND MY S1A products against L2 products from HY2 scatterometer for the four CMEMS regions: a) ARC (Arctic), b) ATL (Atlantic), c) BAL (Baltic) and d) MED (Mediterranean). The 1D histograms of SAR and HY2 wind speeds are also depicted in grey.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. It includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality (Table 9).

The system version in this document corresponds to the "system_version" also described in the Product User Manual (PUM [PUM 012_10X]). It merges information about the version of the processor L2->L3 ("processor version"), the version of L2 OCN products from ESA, the file format version (i.e., "format_version") and any updated of the WIND_XXX_PHY_HR_L3_TT_012_10X products.

Table 19: Main changes for the HR-WIND products.

Date	Change/Event description	System version	Other
June 2024	EIS for NRT – First production	1.0 (NRT only)	Processing version: 1.0 Format version: 1.0
November 2024	EIS for MY - Change of the processing chain	1.1.1 (MY) 1.1.0 (NRT)	Processing version: 1.1 for both NRT and MY Change of the format version: 1.1 for MY (still 1.0 for NRT)
June 2025	EIS for NRT: adding ancillary variables EIS for MY: Backward Temporal Extension (from 27/06/2019 to 27/06/2021) + addition of S1B datasets	1.2	Processing version and format version: 1.2
March 2025	EIS for NRT – Addition of S1C datasets	1.3 (NRT)	Processing version and format version: 1.3 (NRT)

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous version and this one. However, no change since the previous version has been recorded.

VII REFERENCES

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ANNEX A – HISTORICAL VALIDATION ON HIGH-WIND SPEED

The validation of the processing chain for high wind speeds has been reported in various publications (Mouche et al., 2017, 2019) and within the CYMS (CYclone Monitoring Service) Project[6]. It mostly relies on comparing SAR C-band data with co-located wind data obtained either from other satellites, such as the L-band passive microwave measurements from the Soil Moisture Active and Passive mission (SMAP), from the Soil Moisture and Ocean Salinity (SMOS), or from airborne Stepped Frequency Microwave Radiometer (SFMR) measurements.

The relevance of using a dual-pol algorithm instead of a co-pol algorithm is demonstrated through comparisons between co-located SAR and SMAP winds (Mouche et al., 2017).

The use of the GMF MS1AHW2 has been validated against SMOS data within the framework of the CYMS ESA project.

In both cases, collocated wind measurements (i.e., at the same place within a 30 min timespan) are compared and statistics such as biases, standard deviations, scatter index are derived.

VII.1 Mouche et al. 2017: Co-pol vs dual-pol for high winds

Mouche et al. (2017) have shown (see Figure A.1) that using the dual-pol algorithm reduces the bias between SAR and SMAP wind speeds by 0.54 m/s (from 1.32 to 0.78 m/s) across all wind ranges compared to using the co-pol algorithm only. The standard deviation is also reduced by 0.76 m/s (from 3.35 to 2.59 m/s).

For winds higher than 30 m/s, the bias is reduced from 11.27 m/s to 3.35 m/s and the standard deviation from 5.64 m/s to 4.85 m/s.

For the dual-pol algorithm, the comparison with SMAP wind speed yields a bias of 0.78 m/s with a standard deviation of 2.59 m/s.

[6] https://www.esa-cyms.org/

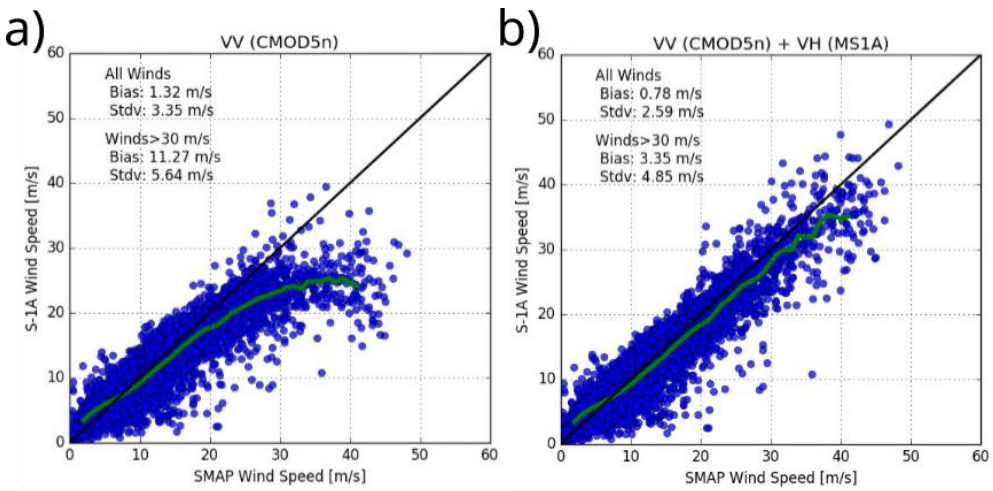

Figure A.1: 40-km SMAP wind versus 40-km SAR wind. SAR winds have been obtained using (a) only VV polarization with CMOD5n GMF, (b) VV polarization with CMOD5n GMF and VH polarization using MS1A GMF (Figure 5 from Mouche et al., 2017)

VII.2 GMF evolution MS1A -> MS1AHW -> MS1AHW2

In Mouche et al. (2019), the SAR NRCS is compared with co-located SFMR wind speeds. Figure A.2.a shows that the MS1A GMF underestimates the NRCS for wind speeds larger than 50 m/s, motivating the development of the MS1AHW GMF in 2019.

Within the framework of the CYMS project, the GMF MS1AHW2 was created to reduce the biases between collocated SMOS, Sentinel-1 SAR and Radarsat-2 SAR wind speeds. This is particularly true for the higher quantiles of wind as shown in Figure A.2.b and Figure A.2.c with a reduced standard deviation from 2.26 m/s to 2.06m/s and a reduced scatter index (SI) from 14.65% to 13.28%.

VII.3 Metrics from reference studies

Table A.1: Performances for various wind inversion schemes. The CMEMS wind inversion is the Dual-pol GMF MS1AHW2

Configuration for SAR wind	Reference wind speed	Bias [m/s] (all winds)	RMSD [m/s] (all winds)	Bias [m/s] (high winds)	RMSD [m/s] (high winds)	Source study
Co-pol	SMAP	1.32	3.35	11.27	5.64	Mouche et al., 2017
Dual-pol GMF MS1A	SMAP	0.78	2.59	3.35	4.85	Mouche et al., 2017
Dual-pol GMF MS1AHW	SMOS	-0.12	2.26	-	-	CYMS validation
Dual-pol GMF MS1AHW2	SMOS	-0.15	2.06	-	-	CYMS validation

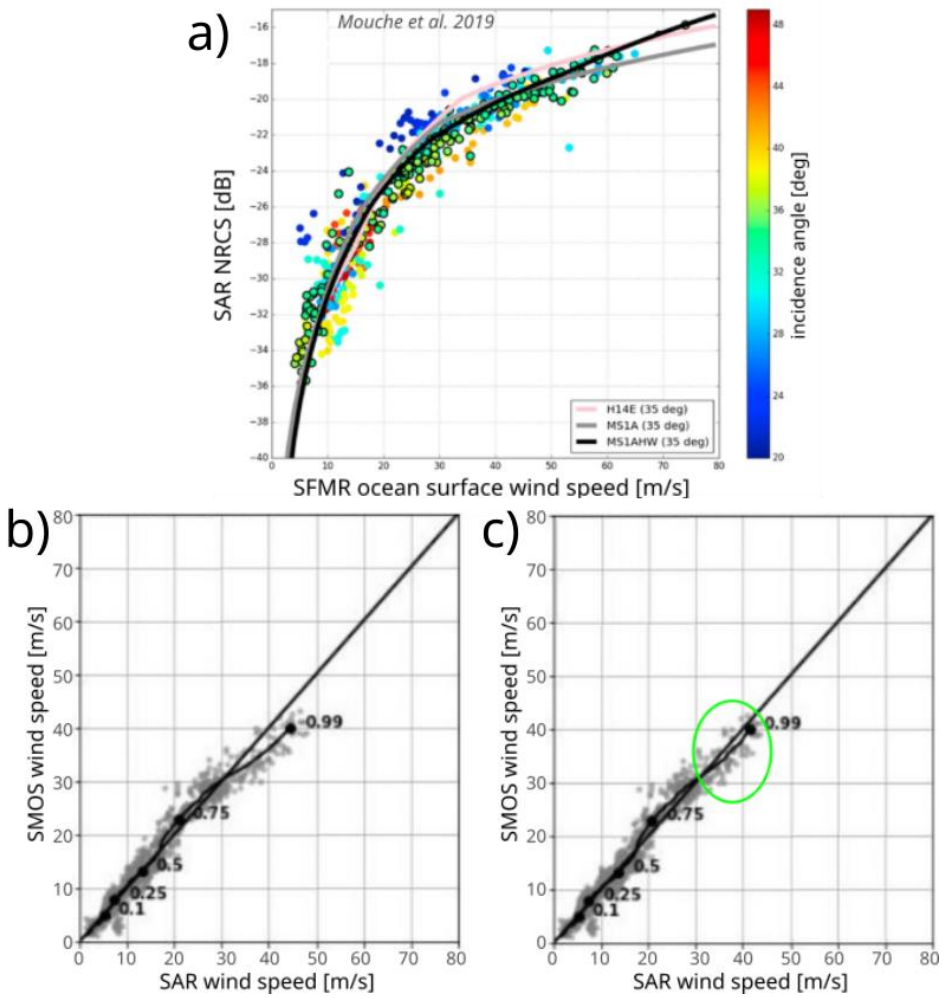

Figure A.2: Evolution of the GMF. (a) Sentinel-1A NRCS measurements in VH polarization as a function of the ocean surface wind speed measured by SFMR and color-coded by incidence angle, solid lines indicate different GMFs, in grey MS1A and in black MS1AHW (taken from Mouche et al., 2019). (b) Co-located SMOS measurements with S1A & S1B SAR wind speed retrieved using MS1AHW GMF, for incidence angles between 30° and 35°. (c) Co-located SMOS measurements with the MS1AHW2 GMF.